



Date: 23/02/25

### Lab Practical #11:

Study and use of Wireshark tool for network traffic and packet analysis.

#### 1. Install Wireshark

Wireshark is pre-installed on Kali Linux, but if you need to install it on other Linux distributions, you can use the following command:

- `sudo apt update`
- `sudo apt install wireshark`

```
(student@student) - [~]
$ sudo apt install wireshark
[sudo] password for student:
The following packages were automatically installed and are no longer required:
aspnetcore-runtime-6.0          llvm-17
aspnetcore-targeting-pack-6.0   llvm-17-runtime
avahi-utils                     medusa
base58                          mingw-w64-common
binutils-mingw-w64-i686         mingw-w64-i686-dev
binutils-mingw-w64-x86-64      mingw-w64-x86-64-dev
blt                             mosquito
comerr-dev                     netstandard-targeting-pack-2.1
dconf-cli                      node-normalize.css
debugedit                      nsis
dnsmas                          nsis-common
dotnet-apphost-pack-6.0        openvas-scanner
dotnet-host                    oracle-instantclient-basic
dotnet-hostfxr-6.0            postgresql
dotnet-runtime-6.0            preview-latex-style
dotnet-runtime-deps-6.0       proj-bin
dotnet-sdk-6.0                proj-data
dotnet-targeting-pack-6.0     python-babel-localedata
dsniff                         python-matplotlib-data
espeak-ng-data                python-odf-doc
```



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|                                           |                     |                     |             |                    |
|-------------------------------------------|---------------------|---------------------|-------------|--------------------|
| llvm-16-runtime                           |                     |                     |             |                    |
| Use 'sudo apt autoremove' to remove them. |                     |                     |             |                    |
| Upgrading:                                | Source              | Destination         | Protocol    | Length Info        |
| clang-17                                  | AzureWaveTec_2c:6   | libstdc++6          | ARP         | 60 Who has 10.20.1 |
| exo-utils                                 | AzureWaveTec_2c:6   | libsvtavlc2         | ARP         | 60 Who has 10.20.1 |
| gcc-14-base                               | fe80::b39e:975d:c   | libtalloc2          | ICMPv6      | 86 Neighbor Solic  |
| gir1.2-gdm-1.0                            | AzureWaveTec_2c:6   | libtdb1             | ARP         | 60 Who has 10.20.1 |
| gir1.2-glib-2.0                           | AzureWaveTec_32:9   | libtevent0t64       | ARP         | 60 Who has 10.20.1 |
| gir1.2-totem-1.0                          | Grandstream_19:8    | libtotem0           | Ethernet II | 628 Ethernet II    |
| gnome-session-bin                         | Grandstream_19:8    | libtsan2            | Ethernet II | 628 Ethernet II    |
| gnome-session-common                      | ce:2f:57            | libubsan1           | ARP         | 60 Who has 10.20.1 |
| gnome-settings-daemon                     | g:186               | libvpl2             | MDNS        | 102 Standard query |
| gnome-settings-daemon-common              | g:186               | libwacom9           | MDNS        | 102 Standard query |
| gnome-shell-common                        | g:186               | libwayland-server0  | ARP         | 60 Who has 10.20.1 |
| gsettings-desktop-schemas                 | d4c:2               | libwbclient0        | ICMPv6      | 86 Neighbor Solic  |
| gvmd-common                               | Sophos_ce:2f:57     | libwireshark-data   | ARP         | 60 Who has 10.20.1 |
| lib32gcc-s1                               | Grandstream_19:8    | libxau-dev          | Ethernet II | 628 Ethernet II    |
| lib32stdc++6                              | 10.20.59.186        | libxau6             | MDNS        | 85 Standard query  |
| libasan8                                  | 10.20.59.186        | libxcb-icccm4       | MDNS        | 102 Standard query |
| libatomic1                                | fe80::87c4:da3b:2   | libxcb-composite1   | ICMPv6      | 86 Neighbor Solic  |
| libcc1-0                                  | Sophos_ce:2f:57     | libxfce4panel-2.0-4 | ARP         | 60 Who has 10.20.1 |
| libclang-common-17-dev                    | Sophos_ce:2f:57     | libxfce4ui-utils    | ARP         | 60 Who has 10.20.1 |
| libclang-cpp17t64                         | 1e:e9:a0            | libxfceconf-0-3     | ICMPv6      | 86 Neighbor Solic  |
| libqt6network6                            | 10:1442:3e85:5      | libxcb-cursor0      |             |                    |
| Suggested packages:                       |                     |                     |             |                    |
| libcudal                                  | Source              | Destination         | Protocol    | Length Info        |
| libnvcuvid1                               | wireshark-doc       | binfmt-support      | ntp         |                    |
| libnvidia-encode1                         | python-pyside2-doc  | bind9               | chrony      |                    |
| libstdc++-14-doc                          | python3.12-venv     | bind9utils          | ARP         | 60 Who has 10.20.1 |
|                                           | python3.12-doc      | ldb-tools           | ARP         | 60 Who has 10.20.1 |
| REMOVING:                                 |                     |                     |             |                    |
| aircrack-ng                               | AzureWaveTec_2c:6   | Broadcast           | ARP         | 60 Who has 10.20.1 |
| bloodhound.py                             | AzureWaveTec_32:9   | Broadcast           | ARP         | 60 Who has 10.20.1 |
| blueman                                   | AzureWaveTec_32:9   | Broadcast           | ARP         | 60 Who has 10.20.1 |
| bully                                     | Sophos_ce:2f:57     | Broadcast           | ARP         | 60 Who has 10.20.1 |
| catfish                                   | 10.20.59.186        | 224.0.0.251         | MDNS        | 102 Standard query |
| certipy-ad                                | 10.20.59.186        | 224.0.0.251         | MDNS        | 102 Standard query |
| chrome-gnome-shell                        | 10:1442:3e85:5      | Broadcast           | ARP         | 60 Who has 10.20.1 |
| commix                                    | 10:1442:3e85:5      | Broadcast           | ARP         | 60 Who has 10.20.1 |
| crackmapexec                              | 10:1442:3e85:5      | Broadcast           | ARP         | 60 Who has 10.20.1 |
| creddump7                                 | 10.20.59.186        | 224.0.0.251         | MDNS        | 85 Standard query  |
| dnsrecon                                  | 10.20.59.186        | 224.0.0.251         | MDNS        | 102 Standard query |
| enum4linux                                | fe80::87c4:da3b:2c5 | ff02::1:ff21:a03    | ICMPv6      | 86 Neighbor Solic  |
| faraday                                   | Sophos_ce:2f:57     | Broadcast           | ARP         | 60 Who has 10.20.1 |
| faraday-agent-dispatcher                  | Sophos_ce:2f:57     | Broadcast           | ARP         | 60 Who has 10.20.1 |
| faraday-cli                               | 1e:e9:a0            | Broadcast           | ARP         | 60 Who has 10.20.1 |
|                                           | fe80::1442:3e85:5e2 | ff02::1:ff21:a03    | ICMPv6      | 86 Neighbor Solic  |

```
system-config-printer-udev
theharvester
totem-plugins
unicorn-magic
urlscan
wafw00f
wapiti
weevely
wfuzz
wifite
winexe
xfce4
xfce4-helpers
xfce4-session
xfce4-settings

Summary:
Upgrading: 177, Installing: 43, Removing: 321, Not Upgrading: 1797
Download size: 239 MB
Freed space: 1917 MB

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 qt6-wayland amd64 6.7.2-5 [143 kB]
```

### 2. Granting Permissions for Capturing Packets

To capture packets, you need to have the proper permissions. On Kali Linux, the following command adds your user to the wireshark group:

- `sudo usermod -aG wireshark $(whoami)`

After running this command, log out and log back in for the changes to take effect.

```
(student@student) - [~]
$ sudo usermod -aG wireshark$(whoami)
Usage: usermod [options] LOGIN

Options:
  -a, --append                append the user to the supplemental GROUPS
                              mentioned by the -G option without removing
                              the user from other groups
  -b, --badname               allow bad names
  -c, --comment COMMENT      new value of the GECOS field
  -d, --home HOME_DIR        new home directory for the user account
  -e, --expiredate EXPIRE_DATE set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE    set password inactive after expiration
                              to INACTIVE
  -g, --gid GROUP             force use GROUP as new primary group
  -G, --groups GROUPS        new list of supplementary GROUPS
  -h, --help                  display this help message and exit
  -l, --login NEW_LOGIN      new value of the login name
  -L, --lock                  lock the user account
  -m, --move-home             move contents of the home directory to the
                              new location (use only with -d)
  -o, --non-unique            allow using duplicate (non-unique) UID
  -p, --password PASSWORD    use encrypted password for the new password
```



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|                              |                                                                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| -h, --help                   | display this help message and exit                                                                                      |
| -l, --login NEW_LOGIN        | new value of the login name                                                                                             |
| -L, --lock                   | lock the user account                                                                                                   |
| -m, --move-home              | move contents of the home directory to the new location (use only with -d)                                              |
| -o, --non-unique             | allow using duplicate (non-unique) UID                                                                                  |
| -p, --password PASSWORD      | use encrypted password for the new password                                                                             |
| -P, --prefix PREFIX_DIR      | prefix directory where are located the /etc/* files                                                                     |
| -r, --remove                 | remove the user from only the supplemental GROUP mentioned by the -G option without removing the user from other groups |
| -R, --root CHROOT_DIR        | directory to chroot into                                                                                                |
| -s, --shell SHELL            | new login shell for the user account                                                                                    |
| -u, --uid UID                | new UID for the user account                                                                                            |
| -U, --unlock                 | unlock the user account                                                                                                 |
| -v, --add-subuids FIRST-LAST | add range of subordinate uids                                                                                           |
| -V, --del-subuids FIRST-LAST | remove range of subordinate uids                                                                                        |
| -w, --add-subgids FIRST-LAST | add range of subordinate gids                                                                                           |
| -W, --del-subgids FIRST-LAST | remove range of subordinate gids                                                                                        |
| -Z, --selinux-user SEUSER    | new SELinux user mapping for the user account                                                                           |
| --selinux-range SERANGE      | new SELinux MLS range for the user account                                                                              |

- **Using Wireshark for Packet Capturing**

1. Launch Wireshark

Open Wireshark from the terminal or application menu:

- wireshark

1. Capturing Live Traffic – Use any website.

2. Applying Capture Filters:

- a. HTTP traffic: http

- i. For HTTP - Inspect request and response headers, URLs, and status codes.

- ii. For HTTPS, observe the handshake process (TLS).





# DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

## Semester 6<sup>th</sup> | Practical Assignment | Cyber Security (23010E004)

Date: 23/02/25

| No. | Time        | Source                  | Destination      | Protocol  | Length | Info                                                                 |
|-----|-------------|-------------------------|------------------|-----------|--------|----------------------------------------------------------------------|
| 1   | 0.000000000 | AzureWaveTec_83:f4:...  | Broadcast        | ARP       | 60     | Who has 10.20.51.192? Tell 10.20.37.9                                |
| 2   | 0.017943253 | AzureWaveTec_2c:6b:...  | Broadcast        | ARP       | 60     | Who has 10.20.53.218? Tell 10.20.45.32                               |
| 3   | 0.017943539 | AzureWaveTec_2c:6b:...  | Broadcast        | ARP       | 60     | Who has 10.20.31.16? Tell 10.20.45.32                                |
| 4   | 0.018277538 | fe80::b39e:975d:c18a... | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:... |
| 5   | 0.018545691 | AzureWaveTec_2c:6b:...  | Broadcast        | ARP       | 60     | Who has 10.20.45.33? Tell 10.20.45.32                                |
| 6   | 0.039590405 | AzureWaveTec_32:9c:...  | Broadcast        | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.56.186                                 |
| 7   | 0.088931818 | Grandstream_15:0b:...   | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 8   | 0.088932050 | Grandstream_15:0b:...   | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 9   | 0.088932068 | Grandstream_15:0b:...   | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 10  | 0.095991704 | Sophos_ce:2f:57         | Broadcast        | ARP       | 60     | Who has 10.20.55.113? Tell 10.20.1.1                                 |
| 11  | 0.104519771 | 10.20.59.186            | 224.0.0.251      | MDNS      | 102    | Standard query 0x965a PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 12  | 0.104519994 | 10.20.59.186            | 224.0.0.251      | MDNS      | 102    | Standard query 0x8ab5 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 13  | 0.107278067 | HP_6a:07:4d             | Broadcast        | ARP       | 60     | Who has 10.20.55.183? Tell 10.20.53.193                              |
| 14  | 0.128005891 | fe80::6ae6:1d4c:25a...  | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 15  | 0.163998827 | Sophos_ce:2f:57         | Broadcast        | ARP       | 60     | Who has 10.20.56.23? Tell 10.20.1.1                                  |
| 16  | 0.205295379 | Grandstream_15:0b:...   | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 17  | 0.205772791 | 10.20.59.186            | 224.0.0.251      | MDNS      | 85     | Standard query 0xb273 PTR Bhargav._dosvc._tcp.local, "QM" question   |
| 18  | 0.207436051 | 10.20.59.186            | 224.0.0.251      | MDNS      | 102    | Standard query 0x32b5 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 19  | 0.272029278 | fe80::87c4:da3b:2c5...  | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from b4:8c:9d:... |
| 20  | 0.319298125 | Sophos_ce:2f:57         | Broadcast        | ARP       | 60     | Who has 10.20.53.218? Tell 10.20.1.1                                 |
| 21  | 0.323979189 | Sophos_ce:2f:57         | Broadcast        | ARP       | 60     | Who has 10.20.57.187? Tell 10.20.1.1                                 |
| 22  | 0.335541563 | 60:ff:9e:1e:3c:88       | Broadcast        | ARP       | 60     | Who has 10.20.39.24? Tell 10.20.59.178                               |
| 23  | 0.386490712 | fe80::1442:3e05:5e2...  | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 24  | 0.386490980 | 60:ff:9e:1e:3c:88       | Broadcast        | ARP       | 60     | Who has 10.20.41.27? Tell 10.20.59.197                               |
| 25  | 0.443162059 | 60:ff:9e:1e:3c:88       | Broadcast        | ARP       | 60     | Who has 10.20.39.30? Tell 10.20.59.196                               |
| 26  | 0.443162618 | 60:ff:9e:1e:3c:88       | Broadcast        | ARP       | 60     | Who has 10.20.39.23? Tell 10.20.59.196                               |
| 27  | 0.471844961 | fe80::2ea2:9cae:571...  | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:... |
| 28  | 0.497925073 | a2:70:1b:bb:19:d0       | Broadcast        | LLC       | 60     | S.P. Func=RN, N(R)=64; DSAP NULL LSAP Individual, SSAP NULL LSAP...  |
| 29  | 0.498635358 | Intel_6b:8c:8a          | Broadcast        | ARP       | 60     | ARP Announcement for 10.20.56.71                                     |
| 30  | 0.530491970 | AzureWaveTec_2c:68:...  | Broadcast        | ARP       | 60     | Who has 10.20.42.33? Tell 10.20.45.19                                |
| 31  | 0.568850592 | 60:ff:9e:1e:3c:f6       | Broadcast        | ARP       | 60     | Who has 10.20.38.17? Tell 10.20.41.31                                |

| No. | Time        | Source                 | Destination            | Protocol  | Length | Info                                                                 |
|-----|-------------|------------------------|------------------------|-----------|--------|----------------------------------------------------------------------|
| 34  | 0.581148196 | DLink_7c:96:20         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 35  | 0.587354123 | AzureWaveTec_2c:68:... | Broadcast              | ARP       | 60     | Who has 10.20.53.218? Tell 10.20.45.19                               |
| 36  | 0.595836804 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59145 → 22222 Len=28                                                 |
| 37  | 0.595837077 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59149 → 22222 Len=28                                                 |
| 38  | 0.595837094 | 10.20.45.24            | 10.20.255.255          | UDP       | 70     | 59147 → 22222 Len=28                                                 |
| 39  | 0.595837113 | 10.20.45.24            | 255.255.255.255        | UDP       | 60     | 59158 → 3289 Len=14                                                  |
| 40  | 0.595837130 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59153 → 22222 Len=28                                                 |
| 41  | 0.595837148 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59150 → 22222 Len=28                                                 |
| 42  | 0.595837166 | 10.20.45.24            | 255.255.255.255        | UDP       | 60     | 59156 → 3289 Len=14                                                  |
| 43  | 0.595837184 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59154 → 22222 Len=28                                                 |
| 44  | 0.595850906 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59146 → 22222 Len=28                                                 |
| 45  | 0.595850930 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59151 → 22222 Len=28                                                 |
| 46  | 0.595850947 | 10.20.45.24            | 255.255.255.255        | UDP       | 70     | 59148 → 22222 Len=28                                                 |
| 47  | 0.595850965 | 10.20.45.24            | 255.255.255.255        | UDP       | 60     | 59155 → 3289 Len=14                                                  |
| 48  | 0.595850982 | 10.20.45.24            | 10.20.255.255          | UDP       | 60     | 59157 → 3289 Len=14                                                  |
| 49  | 0.595850999 | 10.20.45.24            | 255.255.255.255        | UDP       | 60     | 59159 → 3289 Len=14                                                  |
| 50  | 0.595851017 | 10.20.45.24            | 10.20.255.255          | UDP       | 70     | 59152 → 22222 Len=28                                                 |
| 51  | 0.638152145 | 60:ff:9e:1e:3b:2c      | Broadcast              | ARP       | 60     | Who has 10.20.39.24? Tell 10.20.59.191                               |
| 52  | 0.638152388 | 60:ff:9e:1e:3b:2c      | Broadcast              | ARP       | 60     | Who has 10.20.39.29? Tell 10.20.59.191                               |
| 53  | 0.654973800 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.52.19? Tell 10.20.1.1                                  |
| 54  | 0.723909119 | DLink_b5:a2:a0         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 55  | 0.734963075 | Grandstream_d4:50:...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 56  | 0.750612375 | Grandstream_d4:50:...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 57  | 0.760671066 | Grandstream_cd:7e:...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 58  | 0.771170232 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.51.11? Tell 10.20.1.1                                  |
| 59  | 0.771170517 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.58.4? Tell 10.20.1.1                                   |
| 60  | 0.787806658 | fe80::da2d:6e4a:7db... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 61  | 0.834338479 | Grandstream_d4:50:...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 62  | 0.893387722 | fe80::bcf3:f11:1589... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 63  | 0.918137325 | AzureWaveTec_83:f4:... | Broadcast              | ARP       | 60     | Who has 10.20.41.21? Tell 10.20.37.9                                 |
| 64  | 0.918154090 | Grandstream_15:0b:...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |

| No. | Time        | Source                  | Destination            | Protocol  | Length | Info                                                                  |
|-----|-------------|-------------------------|------------------------|-----------|--------|-----------------------------------------------------------------------|
| 190 | 2.447820405 | fe80::b7ba:2905:6e4...  | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:...  |
| 191 | 2.476278934 | fe80::18b1:7680:54f...  | ff02::1:2              | DHCPv6    | 120    | Information-request XID: 0xecfcf3 CID: 000100012d80fa162c58b947b8...  |
| 192 | 2.479364962 | AzureWaveTec_2c:6b:...  | Broadcast              | ARP       | 60     | Who has 10.20.55.58? Tell 10.20.45.32                                 |
| 193 | 2.491160052 | Intel_6b:8c:8a          | Broadcast              | ARP       | 60     | ARP Announcement for 10.20.56.71                                      |
| 194 | 2.498834387 | Sophos_ce:2f:57         | Broadcast              | ARP       | 60     | Who has 10.20.31.10? Tell 10.20.1.1                                   |
| 195 | 2.531465258 | Sophos_ce:2f:57         | Broadcast              | ARP       | 60     | Who has 10.20.26.23? Tell 10.20.1.1                                   |
| 196 | 2.534846160 | 98.82.156.107           | 10.20.51.21            | TCP       | 66     | [TCP ACKed unseen segment] 443 → 44160 [ACK] Seq=1 Ack=2 Win=502 ...  |
| 197 | 2.539103265 | AzureWaveTec_2c:68:...  | Broadcast              | ARP       | 60     | Who has 10.20.53.218? Tell 10.20.45.19                                |
| 198 | 2.540482941 | fe80::b39e:975d:c18a... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::a034:c69e:7e42:1f24 from 1c:ce:51:... |
| 199 | 2.581406583 | DLink_7c:96:20          | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                                |
| 200 | 2.611092748 | fe80::bcf3:f11:1589...  | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:...  |
| 201 | 2.614704323 | fe80::194f:49c1:549...  | ff02::1                | ICMPv6    | 86     | Neighbor Advertisement fe80::194f:49c1:549c:605c (ovr) is at cc:4...  |
| 202 | 2.628756782 | fe80::194f:49c1:549...  | ff02::2                | ICMPv6    | 70     | Router Solicitation from cc:47:40:83:f6:fd                            |
| 203 | 2.635048279 | AzureWaveTec_2c:6b:...  | Broadcast              | ARP       | 60     | Who has 10.20.59.65? Tell 10.20.45.32                                 |
| 204 | 2.654844455 | Sophos_ce:2f:57         | Broadcast              | ARP       | 60     | Who has 10.20.58.73? Tell 10.20.1.1                                   |
| 205 | 2.671886370 | AzureWaveTec_82:ba:...  | Broadcast              | ARP       | 60     | Who has 10.20.50.99? Tell 10.20.51.48                                 |
| 206 | 2.686808159 | Sophos_ce:2f:57         | Broadcast              | ARP       | 60     | Who has 10.20.52.19? Tell 10.20.1.1                                   |
| 207 | 2.694279813 | AzureWaveTec_3c:a6:...  | Broadcast              | ARP       | 60     | Who has 10.20.36.11? Tell 10.20.59.26                                 |
| 208 | 2.714832302 | 10.20.59.65             | 224.0.0.251            | MDNS      | 79     | Standard query 0xae6d PTR _arduino._tcp.local, "QM" question          |
| 209 | 2.720524014 | DLink_b5:a2:a0          | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                                |
| 210 | 2.758956230 | CloudNetwork_77:61:...  | Broadcast              | ARP       | 60     | Who has 10.20.32.7? Tell 10.20.57.214                                 |
| 211 | 2.759592760 | Grandstream_d4:50:...   | Broadcast              | Ethern... | 628    | Ethernet II                                                           |
| 212 | 2.763323507 | 60:ff:9e:1e:3b:42       | Broadcast              | ARP       | 60     | Who has 10.20.38.122? Tell 10.20.41.8                                 |
| 213 | 2.763323816 | 60:ff:9e:1e:3b:42       | Broadcast              | ARP       | 60     | Who has 10.20.38.24? Tell 10.20.41.8                                  |
| 214 | 2.768875894 | 10.20.59.99             | 10.20.255.255          | NBNS      | 110    | Registration NB HP_11<20>                                             |
| 215 | 2.768875967 | AzureWaveTec_83:f6:...  | Broadcast              | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.50.99                                   |
| 216 | 2.772434099 | fe80::da2d:6e4a:7db...  | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:...  |
| 217 | 2.779833853 | 10.20.59.99             | 10.20.255.255          | NBNS      | 110    | Registration NB HP_11<00>                                             |
| 218 | 2.795238389 | 10.20.59.99             | 10.20.255.255          | NBNS      | 110    | Registration NB WORKGROUP<00>                                         |
| 219 | 2.803168370 | Sophos_ce:2f:57         | Broadcast              | ARP       | 60     | Who has 10.20.58.188? Tell 10.20.1.1                                  |
| 220 | 2.898160252 | 10.20.60.7              | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                      |



# DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

## Semester 6<sup>th</sup> | Practical Assignment | Cyber Security (23010E004)

Date: 23/02/25

| No. | Time        | Source                 | Destination            | Protocol  | Length | Info                                                                  |
|-----|-------------|------------------------|------------------------|-----------|--------|-----------------------------------------------------------------------|
| 1   | 0.099000000 | 8a:38:57:66:4a:52      | Broadcast              | ARP       | 60     | Who has 10.20.44.37? Tell 10.20.42.27                                 |
| 2   | 0.099005491 | 60:ff:9e:1e:3e:48      | Broadcast              | ARP       | 60     | Who has 10.20.41.21? Tell 10.20.41.14                                 |
| 3   | 0.016997192 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                           |
| 4   | 0.017041246 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                           |
| 5   | 0.050003657 | Intel_6b:8c:8a         | Broadcast              | ARP       | 60     | Who has 10.20.56.71? (ARP Probe)                                      |
| 6   | 0.050270752 | fe80::68fb:5768:517... | ff02::1                | ICMPv6    | 86     | Neighbor Advertisement fe80::68fb:5768:517:90d2 (ovr) is at e4:b3...  |
| 7   | 0.067969012 | Intel_6b:8c:8a         | Broadcast              | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.56.71                                   |
| 8   | 0.077432526 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.31.16? Tell 10.20.1.1                                   |
| 9   | 0.082676234 | fe80::68fb:5768:517... | ff02::1:3              | LLMNR     | 86     | Standard query 0xb1b7 ANY Vishal                                      |
| 10  | 0.082676466 | 10.20.56.71            | 224.0.0.252            | LLMNR     | 66     | Standard query 0xb1b7 ANY Vishal                                      |
| 11  | 0.084850816 | fe80::797d:c7a6:b6a... | ff02::2                | ICMPv6    | 70     | Router Solicitation from 08:11:96:18:d0:00                            |
| 12  | 0.098684056 | AzureWaveTec_2c:68:... | Broadcast              | ARP       | 60     | Who has 10.20.58.44? Tell 10.20.45.19                                 |
| 13  | 0.102485480 | 54.151.166.244         | 10.20.51.21            | TCP       | 66     | 443 → 51060 [ACK] Seq=1 Ack=1 Win=13 Len=0 TSval=3244386323 TSecr...  |
| 14  | 0.102498041 | 10.20.51.21            | 54.151.166.244         | TCP       | 66     | [TCP ACKed unseen segment] 51060 → 443 [ACK] Seq=1 Ack=2 Win=249 ...  |
| 15  | 0.132948658 | 10.20.51.21            | 142.251.42.99          | TCP       | 66     | 39890 → 80 [ACK] Seq=1 Ack=1 Win=249 Len=0 TSval=3286201076 TSecr...  |
| 16  | 0.132961752 | 10.20.51.21            | 173.194.14.9           | TCP       | 66     | 46072 → 443 [ACK] Seq=1 Ack=1 Win=25904 Len=0 TSval=2419431813 TS...  |
| 17  | 0.144800460 | 142.251.42.99          | 10.20.51.21            | TCP       | 66     | [TCP ACKed unseen segment] 80 → 39890 [ACK] Seq=1 Ack=2 Win=1050 ...  |
| 18  | 0.147747554 | 173.194.14.9           | 10.20.51.21            | TCP       | 66     | [TCP ACKed unseen segment] 443 → 46072 [ACK] Seq=1 Ack=2 Win=1050 ... |
| 19  | 0.154600563 | Intel_d8:27:50         | Broadcast              | ARP       | 60     | Who has 10.20.45.2? Tell 10.20.59.134                                 |
| 20  | 0.154600802 | Intel_d8:27:50         | Broadcast              | ARP       | 60     | Who has 10.20.55.67? Tell 10.20.59.134                                |
| 21  | 0.166416740 | fe80::cc7d:fd78:bea... | ff02::1:3              | LLMNR     | 95     | Standard query 0x86dc A desktop-p6q8s1l                               |
| 22  | 0.166416966 | 10.20.51.218           | 224.0.0.252            | LLMNR     | 75     | Standard query 0x86dc A desktop-p6q8s1l                               |
| 23  | 0.167057311 | fe80::cc7d:fd78:bea... | ff02::1:3              | LLMNR     | 95     | Standard query 0xff40 AAAA desktop-p6q8s1l                            |
| 24  | 0.167057540 | 10.20.51.218           | 224.0.0.252            | LLMNR     | 75     | Standard query 0xff40 AAAA desktop-p6q8s1l                            |
| 25  | 0.374967022 | AzureWaveTec_a4:30:... | Broadcast              | ARP       | 60     | Who has 10.20.31.32? Tell 10.20.58.63                                 |
| 26  | 0.427272984 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.52.19? Tell 10.20.1.1                                   |
| 27  | 0.442996664 | fe80::2ea2:9cae:571... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:...  |
| 28  | 0.462524197 | fe80::da2d:6e4a:7db... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:...  |
| 29  | 0.512613279 | GrandstreamN_cd:7e:... | Broadcast              | Ethern... | 628    | Ethernet II                                                           |
| 30  | 0.518732539 | fe80::268:ebff:fec0... | fe80::797d:c7a6:b6a... | ICMPv6    | 86     | Neighbor Solicitation for fe80::797d:c7a6:b6a2:5e39 from 00:68:eb:... |
| 31  | 0.523999143 | fe80::797d:c7a6:b6a... | fe80::268:ebff:fec0... | ICMPv6    | 86     | Neighbor Advertisement fe80::797d:c7a6:b6a2:5e39 (sol, ovr) is at...  |

| No. | Time        | Source                 | Destination            | Protocol  | Length | Info                                                                 |
|-----|-------------|------------------------|------------------------|-----------|--------|----------------------------------------------------------------------|
| 55  | 0.912036420 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 56  | 0.939681107 | 60:ff:9e:1e:e7:a4      | Broadcast              | ARP       | 60     | Who has 10.20.38.122? Tell 10.20.41.16                               |
| 57  | 0.962315835 | AzureWaveTec_83:f6:... | Broadcast              | ARP       | 60     | Who has 169.254.99.246? Tell 10.20.50.99                             |
| 58  | 0.973617113 | 10.20.60.5             | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 59  | 0.974408636 | 10.20.60.4             | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 60  | 0.976899555 | 10.20.60.10            | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 61  | 0.976899840 | 10.20.60.2             | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 62  | 1.019997147 | fe80::2ea2:9cae:571... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:... |
| 63  | 1.030748733 | fe80::1d95:4973:4ec... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 64  | 1.050842405 | Intel_6b:8c:8a         | Broadcast              | ARP       | 60     | Who has 10.20.56.71? (ARP Probe)                                     |
| 65  | 1.097762511 | AzureWaveTec_2c:68:... | Broadcast              | ARP       | 60     | Who has 10.20.58.44? Tell 10.20.45.19                                |
| 66  | 1.139380644 | DLink_7c:96:20         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 67  | 1.151244829 | Intel_d8:27:50         | Broadcast              | ARP       | 60     | Who has 10.20.55.67? Tell 10.20.59.134                               |
| 68  | 1.151245054 | Intel_d8:27:50         | Broadcast              | ARP       | 60     | Who has 10.20.45.2? Tell 10.20.59.134                                |
| 69  | 1.215615050 | fe80::bcf3:f11:589...  | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 70  | 1.277271455 | 10.20.57.171           | 224.0.0.251            | MDNS      | 424    | Standard query response 0x0000 PTR HP_25_dosvc._tcp.local SRV 0 ...  |
| 71  | 1.277272145 | LiteonTechno_d9:40:... | Broadcast              | ARP       | 60     | Who has 169.254.169.254? Tell 10.20.50.49                            |
| 72  | 1.277704007 | 10.20.57.171           | 224.0.0.251            | MDNS      | 83     | Standard query 0x0000 ANY HP_25_dosvc._tcp.local, "QM" question      |
| 73  | 1.293676280 | DLink_b8:a2:a0         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 74  | 1.307367261 | c2:f5:5c:b5:fc:50      | Broadcast              | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.53.84                                  |
| 75  | 1.330594706 | fe80::da2d:6e4a:7db... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 76  | 1.367419729 | AzureWaveTec_a4:30:... | Broadcast              | ARP       | 60     | Who has 10.20.31.32? Tell 10.20.58.63                                |
| 77  | 1.382232022 | 10.20.59.232           | 224.0.0.251            | MDNS      | 79     | Standard query 0x7eef PTR_arduino._tcp.local, "QM" question          |
| 78  | 1.427319337 | AzureWaveTec_2b:cc:... | Broadcast              | ARP       | 60     | Who has 10.20.45.12? Tell 10.20.42.20                                |
| 79  | 1.454527519 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.52.19? Tell 10.20.1.1                                  |
| 80  | 1.513261600 | GrandstreamN_cd:7f:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 81  | 1.513412545 | GrandstreamN_d4:50:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 82  | 1.539166242 | 10.20.57.171           | 224.0.0.251            | MDNS      | 83     | Standard query 0x0000 ANY HP_25_dosvc._tcp.local, "QM" question      |
| 83  | 1.560596377 | fe80::1d95:4973:4ec... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 84  | 1.597782978 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 85  | 1.597783232 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |

| No. | Time        | Source                 | Destination            | Protocol | Length | Info                                                                 |
|-----|-------------|------------------------|------------------------|----------|--------|----------------------------------------------------------------------|
| 124 | 2.155398803 | fe80::bcf3:f11:589...  | ff02::1:ff21:a03       | ICMPv6   | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 125 | 2.195384841 | da:70:86:ec:67:b3      | Broadcast              | LLC      | 60     | S P; func=RNR, N(R)=64; DSAP NULL LSAP Individual, SSAP NULL LSAP... |
| 126 | 2.206791247 | da:70:86:ec:67:b3      | GrandstreamN_15:0b:... | EAPOL    | 135    | Key (Message 2 of 4)                                                 |
| 127 | 2.214811683 | da:70:86:ec:67:b3      | GrandstreamN_15:0b:... | EAPOL    | 113    | Key (Message 4 of 4)                                                 |
| 128 | 2.267931541 | ::                     | ff02::1:ff63:b1e1      | ICMPv6   | 86     | Neighbor Solicitation for fe80::1004:d2c4:263:b1e1                   |
| 129 | 2.278032386 | 0.0.0.0                | 255.255.255.255        | DHCP     | 342    | DHCP Request - Transaction ID 0xe648ec                               |
| 130 | 2.278827840 | fe80::1004:d2c4:263... | ff02::2                | ICMPv6   | 62     | Router Solicitation                                                  |
| 131 | 2.284168498 | da:70:86:ec:67:b3      | Broadcast              | ARP      | 60     | ARP Announcement for 10.20.51.97                                     |
| 132 | 2.319701255 | c2:f5:5c:b5:fc:50      | Broadcast              | ARP      | 64     | Gratuitous ARP for 10.20.53.84 (Request)                             |
| 133 | 2.334953149 | fe80::da2d:6e4a:7db... | ff02::1:ff21:a03       | ICMPv6   | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 134 | 2.384754886 | AzureWaveTec_fb:01:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.59.14                               |
| 135 | 2.395201092 | CloudNetwork_d3:f1:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.52.76                               |
| 136 | 2.395201318 | AzureWaveTec_3c:a6:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.59.26                               |
| 137 | 2.395201335 | CloudNetwork_55:65:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.53.126                              |
| 138 | 2.395201353 | 56:fb:62:a6:3b:4b      | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.53.189                              |
| 139 | 2.395577678 | AzureWaveTec_9e:47:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.59.144                              |
| 140 | 2.395577944 | AzureWaveTec_e7:11:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.60.48                               |
| 141 | 2.397489964 | Intel_c5:7a:1f         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.59.2                                |
| 142 | 2.401673780 | CloudNetwork_98:7e:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.59.154                              |
| 143 | 2.409144520 | Intel_ed:af:30         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.57.61                               |
| 144 | 2.429223105 | Intel_e9:63:90         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.51.250                              |
| 145 | 2.432941489 | AzureWaveTec_17:1b:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.57.87                               |
| 146 | 2.433241847 | AzureWaveTec_02:37:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.54.231                              |
| 147 | 2.434257094 | Intel_0c:20:42         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.53.1                                |
| 148 | 2.434257320 | AzureWaveTec_64:49:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.52.171                              |
| 149 | 2.434257336 | AzureWaveTec_2b:cc:... | Broadcast              | ARP      | 60     | Who has 10.20.45.12? Tell 10.20.42.20                                |
| 150 | 2.455028507 | 10.20.59.137           | 224.0.0.251            | MDNS     | 79     | Standard query 0x6887 PTR_arduino._tcp.local, "QM" question          |
| 151 | 2.459346398 | Intel_a9:8a:fd         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.54.212                              |
| 152 | 2.459914986 | Intel_c1:dd:83         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.55.185                              |
| 153 | 2.462902760 | LiteonTechno_71:8f:... | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.53.103                              |
| 154 | 2.462903045 | Intel_12:44:fe         | Broadcast              | ARP      | 60     | Who has 10.20.51.206? Tell 10.20.55.49                               |



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| No. | Time         | Source                 | Destination      | Protocol    | Length | Info                                                                   |
|-----|--------------|------------------------|------------------|-------------|--------|------------------------------------------------------------------------|
| 601 | 9.760352758  | 10.20.59.137           | 224.0.0.251      | MDNS        | 102    | Standard query 0x8f97 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...   |
| 602 | 9.760352986  | 10.20.59.137           | 224.0.0.251      | MDNS        | 102    | Standard query 0x8df2 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...   |
| 603 | 9.767221673  | 10.20.59.232           | 224.0.0.251      | MDNS        | 102    | Standard query 0x8681 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...   |
| 604 | 9.767802322  | 10.20.59.232           | 224.0.0.251      | MDNS        | 102    | Standard query 0xfba3 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...   |
| 605 | 9.794571828  | 10.20.60.15            | 224.0.0.120      | IPv4        | 208    | Unassigned (153)                                                       |
| 606 | 9.796482264  | 10.20.60.9             | 224.0.0.120      | IPv4        | 208    | Unassigned (153)                                                       |
| 607 | 9.897345038  | AzureWaveTec_2c:d0:... | Broadcast        | ARP         | 60     | Who has 10.20.45.12? Tell 10.20.45.23                                  |
| 608 | 9.908271277  | Intel_c5:7a:1f         | Broadcast        | ARP         | 60     | Who has 10.20.43.39? Tell 10.20.59.2                                   |
| 609 | 9.919340105  | GrandstreamN_15:0b:... | Broadcast        | Ethernet... | 628    | Ethernet II                                                            |
| 610 | 9.993023475  | MicroStarINT_03:7f:... | Broadcast        | ARP         | 60     | Who has 10.20.26.23? Tell 10.20.26.46                                  |
| 611 | 10.025944922 | GrandstreamN_15:0b:... | Broadcast        | Ethernet... | 628    | Ethernet II                                                            |
| 612 | 10.070137665 | HP_6a:07:75            | Broadcast        | ARP         | 60     | Who has 169.254.169.254? Tell 10.20.56.95                              |
| 613 | 10.070137896 | 60:ff:9e:1e:e7:fc      | Broadcast        | ARP         | 60     | Who has 10.20.38.40? Tell 10.20.41.20                                  |
| 614 | 10.071379433 | 60:ff:9e:1e:e7:fc      | Broadcast        | ARP         | 60     | Who has 10.20.38.122? Tell 10.20.41.20                                 |
| 615 | 10.158602053 | Sophos_ce:2f:57        | Broadcast        | ARP         | 60     | Who has 10.20.52.19? Tell 10.20.1.1                                    |
| 616 | 10.281168923 | Intel_c5:7a:1f         | Broadcast        | ARP         | 60     | Who has 10.20.57.164? Tell 10.20.59.2                                  |
| 617 | 10.321559821 | 10.20.60.22            | 255.255.255.255  | UDP         | 931    | 46328 -> 29810 Len=889                                                 |
| 618 | 10.372372091 | 10.20.59.180           | 224.0.0.251      | MDNS        | 84     | Standard query 0xfe67 PTR Nilesch._dosvc._tcp.local, "QM" question     |
| 619 | 10.372372357 | 10.20.55.220           | 224.0.0.251      | MDNS        | 84     | Standard query 0x68d7 PTR Nilesch._dosvc._tcp.local, "QM" question     |
| 620 | 10.372848192 | 10.20.51.21            | 173.194.14.9     | TCP         | 66     | [TCP Dup ACK 16#1] 46072 -> 443 [ACK] Seq=1 Ack=1 Win=25904 Len=0 T... |
| 621 | 10.372854314 | 10.20.51.21            | 142.251.42.99    | TCP         | 66     | [TCP Dup ACK 15#1] 39090 -> 80 [ACK] Seq=1 Ack=1 Win=249 Len=0 TSV...  |
| 622 | 10.376133947 | 10.20.59.139           | 224.0.0.251      | MDNS        | 84     | Standard query 0x8eac PTR Nilesch._dosvc._tcp.local, "QM" question     |
| 623 | 10.378504227 | 10.20.59.137           | 224.0.0.251      | MDNS        | 84     | Standard query 0x9f0b PTR Nilesch._dosvc._tcp.local, "QM" question     |
| 624 | 10.378504458 | 10.20.59.232           | 224.0.0.251      | MDNS        | 84     | Standard query 0x100b PTR Nilesch._dosvc._tcp.local, "QM" question     |
| 625 | 10.380505605 | 142.251.42.99          | 10.20.51.21      | TCP         | 66     | [TCP Dup ACK 17#1] 80 -> 39890 [ACK] Seq=1 Ack=2 Win=1050 Len=0 TS...  |
| 626 | 10.387935587 | 173.194.14.9           | 10.20.51.21      | TCP         | 66     | [TCP Dup ACK 18#1] 443 -> 46072 [ACK] Seq=1 Ack=2 Win=1050 Len=0 TS... |
| 627 | 10.454859889 | Intel_c5:7a:1f         | Broadcast        | ARP         | 60     | Who has 10.20.31.34? Tell 10.20.59.2                                   |
| 628 | 10.520926471 | GrandstreamN_3d:1b:... | Broadcast        | Ethernet... | 628    | Ethernet II                                                            |
| 629 | 10.520926881 | GrandstreamN_3d:1b:... | Broadcast        | Ethernet... | 628    | Ethernet II                                                            |
| 630 | 10.520926899 | GrandstreamN_d4:50:... | Broadcast        | Ethernet... | 628    | Ethernet II                                                            |
| 631 | 10.55627579  | fe80::bcf3:f11:1589... | ff02::1:ff21:a03 | ICMPv6      | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:...   |

### b. DNS requests: dns

i. Observe how domain names are translated into IP addresses.

ii. Identify DNS request and response packets.

| No. | Time        | Source                 | Destination            | Protocol    | Length | Info                                                                 |
|-----|-------------|------------------------|------------------------|-------------|--------|----------------------------------------------------------------------|
| 1   | 0.000000000 | AzureWaveTec_2b:d0:... | Broadcast              | ARP         | 60     | Who has 10.20.43.22? Tell 10.20.45.18                                |
| 2   | 0.000129226 | AzureWaveTec_2b:d0:... | Broadcast              | ARP         | 60     | Who has 10.20.54.217? Tell 10.20.45.18                               |
| 3   | 0.091316158 | fe80::87cd:da3b:2c5... | ff02::1:ff21:a03       | ICMPv6      | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from b4:8c:9d:... |
| 4   | 0.147424776 | 10.20.59.186           | 224.0.0.251            | MDNS        | 102    | Standard query 0xa1f7 PTR eed53bf1-b007-45b5-a497-116c9e0631e.lo...  |
| 5   | 0.147425010 | 10.20.59.186           | 224.0.0.251            | MDNS        | 102    | Standard query 0x5459 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 6   | 0.160724864 | GrandstreamN_3d:1a:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 7   | 0.166024542 | GrandstreamN_cd:7f:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 8   | 0.167962542 | 60:ff:9e:1e:e7:fc      | Broadcast              | ARP         | 60     | Who has 10.20.39.3? Tell 10.20.41.20                                 |
| 9   | 0.170721133 | GrandstreamN_3d:1b:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 10  | 0.170811049 | GrandstreamN_3d:1b:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 11  | 0.250864950 | 10.20.59.186           | 224.0.0.251            | MDNS        | 102    | Standard query 0x3888 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 12  | 0.250759571 | GrandstreamN_15:0b:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 13  | 0.423783969 | GrandstreamN_3d:1b:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 14  | 0.492619373 | 10.20.60.4             | 224.0.0.120            | IPv4        | 208    | Unassigned (153)                                                     |
| 15  | 0.493982194 | 10.20.60.5             | 224.0.0.120            | IPv4        | 208    | Unassigned (153)                                                     |
| 16  | 0.495524792 | 10.20.60.2             | 224.0.0.120            | IPv4        | 208    | Unassigned (153)                                                     |
| 17  | 0.496504971 | 10.20.60.10            | 224.0.0.120            | IPv4        | 208    | Unassigned (153)                                                     |
| 18  | 0.520167802 | fe80::94ac:a223:c2f... | ff02::1:3              | LLMNR       | 88     | Standard query 0x5d59 A 02_14_21                                     |
| 19  | 0.520168189 | 10.20.51.131           | 224.0.0.252            | LLMNR       | 68     | Standard query 0x5d59 A 02_14_21                                     |
| 20  | 0.520168206 | fe80::94ac:a223:c2f... | ff02::1:3              | LLMNR       | 88     | Standard query 0x12ea AAAA 02_14_21                                  |
| 21  | 0.520168225 | 10.20.51.131           | 224.0.0.252            | LLMNR       | 68     | Standard query 0x12ea AAAA 02_14_21                                  |
| 22  | 0.568717957 | Dlink_7c:96:20         | Ethernet-Configurat... | LOOP        | 60     | Unknown function (256)                                               |
| 23  | 0.601741556 | 60:ff:9e:1e:e9:98      | Broadcast              | ARP         | 60     | Who has 10.20.39.3? Tell 10.20.41.39                                 |
| 24  | 0.617872242 | GrandstreamN_15:0b:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 25  | 0.617899413 | GrandstreamN_15:0b:... | Broadcast              | Ethernet... | 628    | Ethernet II                                                          |
| 26  | 0.636940740 | fe80::7215:2a71:611... | ff02::1:ff09:3ac1      | ICMPv6      | 86     | Neighbor Solicitation for fe80::20f6:9668:8a09:3ac1 from b8:1e:a4... |

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0  
Ethernet II, Src: AzureWaveTec\_2b:d0:18 (1c:ce:51:2b:d0:18), Dst: Broadcast (ff:ff:ff:ff:ff:ff)  
Address Resolution Protocol (request)





# DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 6<sup>th</sup> | Practical Assignment | Cyber Security (23010E004)

Date: 23/02/25

| Apply a display filter                                                                          |             |                        |                  |           |        |                                                                      |
|-------------------------------------------------------------------------------------------------|-------------|------------------------|------------------|-----------|--------|----------------------------------------------------------------------|
| No.                                                                                             | Time        | Source                 | Destination      | Protocol  | Length | Info                                                                 |
| 138                                                                                             | 2.992725553 | fe80::1d95:4973:4ec... | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 139                                                                                             | 3.011983876 | AzureWaveTec_2c:68:... | Broadcast        | ARP       | 60     | Who has 10.20.36.37? Tell 10.20.45.19                                |
| 140                                                                                             | 3.013831514 | 10.20.59.65            | 224.0.0.251      | MDNS      | 83     | Standard query 0x968a PTR VIJAY_dosvc_tcp.local, "QM" question       |
| 141                                                                                             | 3.040545304 | GrandstreamN_d4:50:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 142                                                                                             | 3.091194230 | HP_6a:07:4d            | Broadcast        | ARP       | 60     | Who has 10.20.2.14? Tell 10.20.53.193                                |
| 143                                                                                             | 3.093936140 | 60:ff:9e:1e:3c:6c      | Broadcast        | ARP       | 60     | Who has 10.20.39.9? Tell 10.20.59.196                                |
| 144                                                                                             | 3.093936371 | 60:ff:9e:1e:3c:6c      | Broadcast        | ARP       | 60     | Who has 10.20.38.40? Tell 10.20.59.196                               |
| 145                                                                                             | 3.106887748 | fe80::6ae6:1d4c:25a... | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 146                                                                                             | 3.117928248 | 10.20.59.65            | 224.0.0.251      | MDNS      | 102    | Standard query 0x25da PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 147                                                                                             | 3.117928595 | 10.20.59.186           | 224.0.0.251      | MDNS      | 102    | Standard query 0xd970 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 148                                                                                             | 3.117928613 | 10.20.59.65            | 224.0.0.251      | MDNS      | 102    | Standard query 0x63f8 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 149                                                                                             | 3.118679866 | 10.20.59.186           | 224.0.0.251      | MDNS      | 102    | Standard query 0x54e8 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 150                                                                                             | 3.126099850 | GrandstreamN_3d:1b:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 151                                                                                             | 3.150154260 | GrandstreamN_d4:50:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 152                                                                                             | 3.162950535 | GrandstreamN_cd:7e:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 153                                                                                             | 3.162975375 | GrandstreamN_cd:7e:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 154                                                                                             | 3.168731919 | GrandstreamN_cd:7f:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 155                                                                                             | 3.223700735 | 10.20.59.65            | 224.0.0.251      | MDNS      | 102    | Standard query 0x04de PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 156                                                                                             | 3.223700961 | 10.20.59.186           | 224.0.0.251      | MDNS      | 102    | Standard query 0x2cfa PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 157                                                                                             | 3.232727024 | AzureWaveTec_2b:d0:... | Broadcast        | ARP       | 60     | Who has 10.20.43.2? Tell 10.20.45.18                                 |
| 158                                                                                             | 3.232727250 | AzureWaveTec_2b:d0:... | Broadcast        | ARP       | 60     | Who has 10.20.54.217? Tell 10.20.45.18                               |
| 159                                                                                             | 3.250547423 | Sophos_ce:2f:57        | Broadcast        | ARP       | 60     | Who has 10.20.58.4? Tell 10.20.1.1                                   |
| 160                                                                                             | 3.264754027 | 92:0e:21:cc:c9:13      | Broadcast        | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.58.205                                 |
| 161                                                                                             | 3.365790600 | fe80::1442:3e85:5e2... | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 162                                                                                             | 3.613031809 | AzureWaveTec_2c:6b:... | Broadcast        | ARP       | 60     | Who has 10.20.31.16? Tell 10.20.45.32                                |
| 163                                                                                             | 3.618992520 | AzureWaveTec_2c:6b:... | Broadcast        | ARP       | 60     | Who has 10.20.31.15? Tell 10.20.45.32                                |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0      |             |                        |                  |           |        |                                                                      |
| Ethernet II, Src: AzureWaveTec_2b:d0:10 (1c:ce:51:2b:d0:10), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |             |                        |                  |           |        |                                                                      |
| Address Resolution Protocol (request)                                                           |             |                        |                  |           |        |                                                                      |

| Apply a display filter                                                                          |             |                        |                  |           |        |                                                                      |
|-------------------------------------------------------------------------------------------------|-------------|------------------------|------------------|-----------|--------|----------------------------------------------------------------------|
| No.                                                                                             | Time        | Source                 | Destination      | Protocol  | Length | Info                                                                 |
| 171                                                                                             | 3.787722330 | AzureWaveTec_e7:dd:... | Broadcast        | ARP       | 60     | Who has 10.20.45.1? Tell 10.20.59.249                                |
| 172                                                                                             | 3.855401960 | 0.0.0.0                | 255.255.255.255  | DHCP      | 342    | DHCP Discover - Transaction ID 0x3fedce56                            |
| 173                                                                                             | 3.855647948 | Sophos_ce:2f:57        | Broadcast        | ARP       | 60     | Who has 10.20.53.18? Tell 10.20.1.1                                  |
| 174                                                                                             | 3.909266883 | 60:ff:9e:1e:3c:6c      | Broadcast        | ARP       | 60     | Who has 10.20.39.9? Tell 10.20.59.196                                |
| 175                                                                                             | 3.909267167 | 60:ff:9e:1e:3c:6c      | Broadcast        | ARP       | 60     | Who has 10.20.38.40? Tell 10.20.59.196                               |
| 176                                                                                             | 3.932320968 | 10.20.60.13            | 224.0.0.120      | IPv4      | 208    | Unassigned (153)                                                     |
| 177                                                                                             | 3.935932536 | 10.20.60.14            | 224.0.0.120      | IPv4      | 208    | Unassigned (153)                                                     |
| 178                                                                                             | 3.938392019 | 10.20.60.19            | 224.0.0.120      | IPv4      | 208    | Unassigned (153)                                                     |
| 179                                                                                             | 3.985740520 | 92:0e:21:cc:c9:13      | Broadcast        | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.58.205                                 |
| 180                                                                                             | 4.009989765 | AzureWaveTec_2c:68:... | Broadcast        | ARP       | 60     | Who has 10.20.36.37? Tell 10.20.45.19                                |
| 181                                                                                             | 4.010557838 | AzureWaveTec_2b:d0:... | Broadcast        | ARP       | 60     | Who has 10.20.54.217? Tell 10.20.45.18                               |
| 182                                                                                             | 4.010558249 | AzureWaveTec_2b:d0:... | Broadcast        | ARP       | 60     | Who has 10.20.43.2? Tell 10.20.45.18                                 |
| 183                                                                                             | 4.085922099 | HP_6a:07:4d            | Broadcast        | ARP       | 60     | Who has 10.20.2.14? Tell 10.20.53.193                                |
| 184                                                                                             | 4.143814433 | 10.20.59.65            | 224.0.0.251      | MDNS      | 102    | Standard query 0x6e71 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 185                                                                                             | 4.143814703 | 10.20.59.186           | 224.0.0.251      | MDNS      | 102    | Standard query 0x78e6 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 186                                                                                             | 4.143814720 | 10.20.59.186           | 224.0.0.251      | MDNS      | 102    | Standard query 0x0e71 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 187                                                                                             | 4.143814738 | 10.20.59.65            | 224.0.0.251      | MDNS      | 102    | Standard query 0x41ab PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 188                                                                                             | 4.181362394 | fe80::94ac:a223:c2f... | ff02::1:3        | LLMNR     | 88     | Standard query 0x896b A 02_14_21                                     |
| 189                                                                                             | 4.181362623 | fe80::94ac:a223:c2f... | ff02::1:3        | LLMNR     | 88     | Standard query 0x667f AAAA 02_14_21                                  |
| 190                                                                                             | 4.181362640 | 10.20.51.131           | 224.0.0.252      | LLMNR     | 68     | Standard query 0x896b A 02_14_21                                     |
| 191                                                                                             | 4.181362658 | 10.20.51.131           | 224.0.0.252      | LLMNR     | 68     | Standard query 0x667f AAAA 02_14_21                                  |
| 192                                                                                             | 4.242948970 | 10.20.59.186           | 224.0.0.251      | MDNS      | 102    | Standard query 0xf610 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 193                                                                                             | 4.242949681 | 10.20.59.65            | 224.0.0.251      | MDNS      | 102    | Standard query 0xd644 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 194                                                                                             | 4.258383369 | fe80::6ae6:1d4c:25a... | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 60:ff:9e:... |
| 195                                                                                             | 4.262681514 | GrandstreamN_15:0b:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 196                                                                                             | 4.267079299 | 10.20.51.21            | 23.193.114.57    | TLSv1.2   | 1408   | Application Data                                                     |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0      |             |                        |                  |           |        |                                                                      |
| Ethernet II, Src: AzureWaveTec_2b:d0:10 (1c:ce:51:2b:d0:10), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |             |                        |                  |           |        |                                                                      |
| Address Resolution Protocol (request)                                                           |             |                        |                  |           |        |                                                                      |

c. TCP packets: tcp

i. Observe the 3-way handshake (SYN, SYN-ACK, ACK).

ii. Identify sequence numbers and acknowledgment numbers.





# DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 6<sup>th</sup> | Practical Assignment | Cyber Security (23010E004)

Date: 23/02/25

| https://wiki.wireshark.org/TCP_3_way_handshaking                                           |              |                        |               |             |        |                                                                      |
|--------------------------------------------------------------------------------------------|--------------|------------------------|---------------|-------------|--------|----------------------------------------------------------------------|
| No.                                                                                        | Time         | Source                 | Destination   | Protocol    | Length | Info                                                                 |
| 35                                                                                         | 0.906302085  | AzureWaveTec_2c:68:... | Broadcast     | ARP         | 60     | Who has 10.20.45.6? Tell 10.20.45.8                                  |
| 34                                                                                         | 0.906301797  | AzureWaveTec_2c:68:... | Broadcast     | ARP         | 60     | Who has 10.20.45.2? Tell 10.20.45.8                                  |
| 32                                                                                         | 0.761293966  | GrandstreamN_15:0b:... | Broadcast     | Ethernet II | 628    | Ethernet II                                                          |
| 31                                                                                         | 0.599995999  | GrandstreamN_d4:50:... | Broadcast     | Ethernet II | 628    | Ethernet II                                                          |
| 27                                                                                         | 0.561346267  | Sophos_ce:2f:57        | Broadcast     | ARP         | 60     | Who has 10.20.58.105? Tell 10.20.1.1                                 |
| 10                                                                                         | 0.146100284  | GrandstreamN_15:0b:... | Broadcast     | Ethernet II | 628    | Ethernet II                                                          |
| 9                                                                                          | 0.145965312  | GrandstreamN_15:0b:... | Broadcast     | Ethernet II | 628    | Ethernet II                                                          |
| 8                                                                                          | 0.107601545  | GrandstreamN_d4:50:... | Broadcast     | Ethernet II | 628    | Ethernet II                                                          |
| 7                                                                                          | 0.106562696  | GrandstreamN_15:0b:... | Broadcast     | Ethernet II | 628    | Ethernet II                                                          |
| 6                                                                                          | 0.066216129  | AzureWaveTec_e7:dd:... | Broadcast     | ARP         | 60     | Who has 10.20.60.238? Tell 10.20.59.249                              |
| 5                                                                                          | 0.057482228  | AzureWaveTec_2c:68:... | Broadcast     | ARP         | 60     | Who has 10.20.45.6? Tell 10.20.45.8                                  |
| 4                                                                                          | 0.057481935  | AzureWaveTec_2c:68:... | Broadcast     | ARP         | 60     | Who has 10.20.45.2? Tell 10.20.45.8                                  |
| 1                                                                                          | 0.000000000  | HP_6a:08:bd            | Broadcast     | ARP         | 60     | Who has 10.20.1.1? Tell 10.20.4.37                                   |
| 1093                                                                                       | 17.753145971 | 10.20.51.21            | 52.25.107.121 | TCP         | 66     | 38040 → 443 [ACK] Seq=79 Ack=2 Win=517 Len=0 TSval=267942615 TSec... |
| 1077                                                                                       | 17.468498866 | 10.20.51.21            | 52.25.107.121 | TCP         | 66     | 38040 → 443 [FIN, ACK] Seq=78 Ack=1 Win=517 Len=0 TSval=267942331... |
| 1076                                                                                       | 17.468492049 | 10.20.51.21            | 52.25.107.121 | TLSv1.2     | 97     | Encrypted Alert                                                      |
| 1073                                                                                       | 17.468139031 | 10.20.51.21            | 52.25.107.121 | TLSv1.2     | 112    | Application Data                                                     |
| 1089                                                                                       | 17.722769634 | 10.20.51.21            | 35.83.144.64  | TCP         | 66     | 54204 → 443 [ACK] Seq=79 Ack=2 Win=249 Len=0 TSval=317254237 TSec... |
| 1075                                                                                       | 17.468291992 | 10.20.51.21            | 35.83.144.64  | TCP         | 66     | 54204 → 443 [FIN, ACK] Seq=78 Ack=1 Win=249 Len=0 TSval=317253982... |
| 1074                                                                                       | 17.468283272 | 10.20.51.21            | 35.83.144.64  | TLSv1.2     | 97     | Encrypted Alert                                                      |
| 1072                                                                                       | 17.468106210 | 10.20.51.21            | 35.83.144.64  | TLSv1.2     | 112    | Application Data                                                     |
| 64                                                                                         | 1.670959778  | 10.20.51.21            | 35.167.38.191 | TCP         | 66     | 41764 → 443 [ACK] Seq=47 Ack=47 Win=249 Len=0 TSval=3786273771 TS... |
| 49                                                                                         | 1.415783520  | 10.20.51.21            | 35.167.38.191 | TLSv1.2     | 112    | Application Data                                                     |
| 1038                                                                                       | 16.720694979 | 10.20.51.21            | 35.166.60.117 | TCP         | 66     | 60762 → 443 [ACK] Seq=79 Ack=2 Win=249 Len=0 TSval=340684358 TSec... |
| 1017                                                                                       | 16.468110046 | 10.20.51.21            | 35.166.60.117 | TCP         | 66     | 60762 → 443 [FIN, ACK] Seq=78 Ack=1 Win=249 Len=0 TSval=340684105... |
| 1016                                                                                       | 16.468102549 | 10.20.51.21            | 35.166.60.117 | TLSv1.2     | 97     | Encrypted Alert                                                      |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0 |              |                        |               |             |        |                                                                      |
| Ethernet II, Src: HP_6a:08:bd (bc:0f:f3:6a:08:bd), Dst: Broadcast (ff:ff:ff:ff:ff:ff)      |              |                        |               |             |        |                                                                      |
| Address Resolution Protocol (request)                                                      |              |                        |               |             |        |                                                                      |

| https://wiki.wireshark.org/TCP_3_way_handshaking                                           |              |              |                 |          |        |                                                                      |
|--------------------------------------------------------------------------------------------|--------------|--------------|-----------------|----------|--------|----------------------------------------------------------------------|
| No.                                                                                        | Time         | Source       | Destination     | Protocol | Length | Info                                                                 |
| 571                                                                                        | 9.131767856  | 10.20.51.21  | 34.117.188.166  | QUIC     | 1399   | Initial, DCID=a9f977d8b81041d4556d0c93c6, SCID=6a20fd, PKN: 0, CR... |
| 1849                                                                                       | 31.601282775 | 10.20.1.1    | 255.255.255.255 | DHCP     | 342    | DHCP ACK - Transaction ID 0xb2f311c                                  |
| 1848                                                                                       | 31.601254965 | 0.0.0.0      | 255.255.255.255 | DHCP     | 352    | DHCP Request - Transaction ID 0xb2f311c                              |
| 1736                                                                                       | 30.025220314 | 0.0.0.0      | 255.255.255.255 | DHCP     | 350    | DHCP Request - Transaction ID 0x8ac57a86                             |
| 1676                                                                                       | 29.011817095 | 0.0.0.0      | 255.255.255.255 | DHCP     | 342    | DHCP Discover - Transaction ID 0x8ac57a86                            |
| 1562                                                                                       | 25.870604678 | 10.20.8.16   | 255.255.255.255 | UDP      | 79     | 59258 → 1124 Len=37                                                  |
| 1533                                                                                       | 25.195570966 | 10.20.60.21  | 255.255.255.255 | UDP      | 936    | 44831 → 29810 Len=894                                                |
| 1513                                                                                       | 24.856482614 | 10.20.8.16   | 255.255.255.255 | UDP      | 60     | 35686 → 3289 Len=15                                                  |
| 1406                                                                                       | 22.383029639 | 10.20.1.1    | 255.255.255.255 | DHCP     | 342    | DHCP Offer - Transaction ID 0x90ebd989                               |
| 1404                                                                                       | 22.311228451 | 10.20.8.16   | 255.255.255.255 | SNMP     | 115    | get-request 1.3.6.1.2.1.1.1.0 1.3.6.1.2.1.1.2.0 1.3.6.1.2.1.2.2.1... |
| 1375                                                                                       | 21.382259944 | 0.0.0.0      | 255.255.255.255 | DHCP     | 342    | DHCP Discover - Transaction ID 0x90ebd989                            |
| 1336                                                                                       | 21.311118650 | 10.20.8.16   | 255.255.255.255 | SNMP     | 115    | get-request 1.3.6.1.2.1.1.1.0 1.3.6.1.2.1.1.2.0 1.3.6.1.2.1.2.2.1... |
| 1115                                                                                       | 18.110353618 | 10.20.1.1    | 255.255.255.255 | DHCP     | 342    | DHCP ACK - Transaction ID 0xec48dd5                                  |
| 1114                                                                                       | 18.110353618 | 0.0.0.0      | 255.255.255.255 | DHCP     | 352    | DHCP Request - Transaction ID 0xec48dd5                              |
| 1043                                                                                       | 16.847313290 | 10.20.1.1    | 255.255.255.255 | DHCP     | 342    | DHCP ACK - Transaction ID 0xec48dd5                                  |
| 1042                                                                                       | 16.847313064 | 0.0.0.0      | 255.255.255.255 | DHCP     | 352    | DHCP Request - Transaction ID 0xec48dd5                              |
| 1019                                                                                       | 16.513432793 | 0.0.0.0      | 255.255.255.255 | DHCP     | 342    | DHCP Inform - Transaction ID 0x7282094a                              |
| 955                                                                                        | 15.162122345 | 10.20.60.21  | 255.255.255.255 | UDP      | 935    | 44831 → 29810 Len=893                                                |
| 948                                                                                        | 15.043192958 | 0.0.0.0      | 255.255.255.255 | DHCP     | 339    | DHCP Request - Transaction ID 0x68a086c                              |
| 880                                                                                        | 14.042777015 | 0.0.0.0      | 255.255.255.255 | DHCP     | 333    | DHCP Discover - Transaction ID 0x68a086c                             |
| 393                                                                                        | 6.663515886  | 10.20.1.1    | 255.255.255.255 | DHCP     | 342    | DHCP Offer - Transaction ID 0xe28f0e8                                |
| 336                                                                                        | 5.662741011  | 0.0.0.0      | 255.255.255.255 | DHCP     | 342    | DHCP Discover - Transaction ID 0xe28f0e8                             |
| 304                                                                                        | 5.137030734  | 10.20.60.21  | 255.255.255.255 | UDP      | 933    | 44831 → 29810 Len=891                                                |
| 259                                                                                        | 4.383140140  | 10.20.51.131 | 255.255.255.255 | UDP      | 82     | 52937 → 1947 Len=40                                                  |
| 1896                                                                                       | 32.626918482 | 10.20.28.4   | 224.0.0.252     | LLMNR    | 68     | Standard query 0x84d2 A 05_18_04                                     |
| 1895                                                                                       | 32.626918465 | 10.20.28.4   | 224.0.0.252     | LLMNR    | 68     | Standard query 0x1459 AAAA 05_18_04                                  |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0 |              |              |                 |          |        |                                                                      |
| Ethernet II, Src: HP_6a:08:bd (bc:0f:f3:6a:08:bd), Dst: Broadcast (ff:ff:ff:ff:ff:ff)      |              |              |                 |          |        |                                                                      |
| Address Resolution Protocol (request)                                                      |              |              |                 |          |        |                                                                      |

| https://wiki.wireshark.org/TCP_3_way_handshaking                                           |              |              |             |          |        |                                                                      |
|--------------------------------------------------------------------------------------------|--------------|--------------|-------------|----------|--------|----------------------------------------------------------------------|
| No.                                                                                        | Time         | Source       | Destination | Protocol | Length | Info                                                                 |
| 1601                                                                                       | 26.587204291 | 10.20.53.193 | 224.0.0.251 | MDNS     | 431    | Standard query response 0x0000 PTR, cache flush HP_8._dosvc._tcp...  |
| 1595                                                                                       | 26.468325656 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x9e21 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 1594                                                                                       | 26.468325639 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0xbfcf PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 1593                                                                                       | 26.468325414 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0xa027 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 1592                                                                                       | 26.320948410 | 10.20.53.193 | 224.0.0.251 | MDNS     | 82     | Standard query 0x0000 ANY HP_8._dosvc._tcp.local, "QM" question      |
| 1571                                                                                       | 26.054896536 | 10.20.53.193 | 224.0.0.251 | MDNS     | 82     | Standard query 0x0000 ANY HP_8._dosvc._tcp.local, "QM" question      |
| 1556                                                                                       | 25.793044112 | 10.20.53.193 | 224.0.0.251 | MDNS     | 82     | Standard query 0x0000 ANY HP_8._dosvc._tcp.local, "QM" question      |
| 1555                                                                                       | 25.790285360 | 10.20.53.193 | 224.0.0.251 | MDNS     | 377    | Standard query response 0x0000 PTR HP_8._dosvc._tcp.local SRV 0 0... |
| 1544                                                                                       | 25.545667796 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x23ba PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 1537                                                                                       | 25.443101117 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x38e6 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 1536                                                                                       | 25.443100844 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x7250 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 1495                                                                                       | 24.521719515 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x8ec1 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 1494                                                                                       | 24.521719498 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x64de PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 1493                                                                                       | 24.521719248 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x104b PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 1472                                                                                       | 23.969051184 | 10.20.8.16   | 224.0.0.251 | MDNS     | 79     | Standard query 0x0000 PTR _scanner._tcp.local, "QM" question         |
| 1451                                                                                       | 23.496871564 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0xae32 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 1450                                                                                       | 23.496871337 | 10.20.59.186 | 224.0.0.251 | MDNS     | 70     | Standard query 0x905d PTR YASH.local, "QM" question                  |
| 1449                                                                                       | 23.496322702 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x5666 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 1448                                                                                       | 23.496322475 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x472d PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 1439                                                                                       | 23.192436853 | 10.20.59.186 | 224.0.0.251 | MDNS     | 86     | Standard query 0xf1f0 PTR H-205-32._dosvc._tcp.local, "QM" questi... |
| 1429                                                                                       | 22.968705695 | 10.20.8.16   | 224.0.0.251 | MDNS     | 79     | Standard query 0x0000 PTR _scanner._tcp.local, "QM" question         |
| 1411                                                                                       | 22.473707931 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x530c PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 1410                                                                                       | 22.473707913 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0x8983 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 1409                                                                                       | 22.473707686 | 10.20.59.186 | 224.0.0.251 | MDNS     | 102    | Standard query 0xe0a5 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 1405                                                                                       | 22.356354399 | 10.20.8.16   | 224.0.0.251 | MDNS     | 87     | Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR _ip... |
| 1389                                                                                       | 21.806849304 | 10.20.59.186 | 224.0.0.251 | MDNS     | 79     | Standard query 0xc719 PTR arduino._tcp.local, "QM" question          |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0 |              |              |             |          |        |                                                                      |
| Ethernet II, Src: HP_6a:08:bd (bc:0f:f3:6a:08:bd), Dst: Broadcast (ff:ff:ff:ff:ff:ff)      |              |              |             |          |        |                                                                      |
| Address Resolution Protocol (request)                                                      |              |              |             |          |        |                                                                      |

Date: 23/02/25

### • Description:-

1. TCB---Transmission Control Block, something like PCB, it stores some significant info like, TCP connection table, the pointer for the sending and receiving buffer, retransmission queue pointer, the current sequence number and acknowledge number and ext.
2. ACK packet could take data content, if not, this packet will not consume SYN number. So the sequence number of the confirm packet is seq=x+1. But in practice, at times, TCP 3-way handshake not only just initiates the connection, but also negotiate some very important parameters. MSS (maximum segment size) negotiation occurs in this steps.

d. ICMP (ping): icmp

i. Send ping requests (ping google.com) and observe request/reply packets.

| No. | Time        | Source                 | Destination       | Protocol  | Length | Info                                                                  |
|-----|-------------|------------------------|-------------------|-----------|--------|-----------------------------------------------------------------------|
| 1   | 0.000000000 | AzureWaveTec_2c:6b:... | Broadcast         | ARP       | 60     | who has 10.20.53.218? Tell 10.20.45.32                                |
| 2   | 0.001513387 | 10.20.60.14            | 224.0.0.120       | IPv4      | 208    | Unassigned (153)                                                      |
| 3   | 0.004002614 | 10.20.60.19            | 224.0.0.120       | IPv4      | 208    | Unassigned (153)                                                      |
| 4   | 0.004046877 | 10.20.60.13            | 224.0.0.120       | IPv4      | 208    | Unassigned (153)                                                      |
| 5   | 0.029972114 | GrandstreamN_15:0b:... | Broadcast         | Ethern... | 628    | Ethernet II                                                           |
| 6   | 0.030591200 | GrandstreamN_15:0b:... | Broadcast         | Ethern... | 628    | Ethernet II                                                           |
| 7   | 0.030712226 | GrandstreamN_15:0b:... | Broadcast         | Ethern... | 628    | Ethernet II                                                           |
| 8   | 0.053601703 | fe80::1cf0:d8eb:706... | ff02::1:ffb5:2e12 | ICMPv6    | 86     | Neighbor Solicitation for fe80::8fc:407b:48b5:2e12 from 3c:a6:f6:...  |
| 9   | 0.053602587 | fe80::1cf0:d8eb:706... | ff02::1:ffc0:8864 | ICMPv6    | 86     | Neighbor Solicitation for fe80::a8ab:b5ff:fec0:8864 from 3c:a6:f6:... |
| 10  | 0.053602606 | fe80::1cf0:d8eb:706... | ff02::1:ff30:166f | ICMPv6    | 86     | Neighbor Solicitation for fe80::1480:61c9:ae30:166f from 3c:a6:f6:... |
| 11  | 0.053602624 | fe80::1cf0:d8eb:706... | ff02::1:ffea:76c5 | ICMPv6    | 86     | Neighbor Solicitation for fe80::1cfa:6a:42ea:76c5 from 3c:a6:f6:4...  |
| 12  | 0.053602643 | fe80::1cf0:d8eb:706... | ff02::1:ffdb:dc03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::c3e:6bf:10db:dc03 from 3c:a6:f6:4...  |
| 13  | 0.053648644 | fe80::1cf0:d8eb:706... | ff02::1:ffb7:711f | ICMPv6    | 86     | Neighbor Solicitation for fe80::1c2a:cba1:f0b7:711f from 3c:a6:f6:... |
| 14  | 0.161781104 | 10.20.59.65            | 224.0.0.251       | MDNS      | 102    | Standard query 0xc076 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...  |
| 15  | 0.163026374 | 10.20.59.65            | 224.0.0.251       | MDNS      | 102    | Standard query 0x95d9 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...  |
| 16  | 0.163026450 | 10.20.59.65            | 224.0.0.251       | MDNS      | 102    | Standard query 0x83b8 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...  |
| 17  | 0.163965736 | 10.20.59.134           | 224.0.0.251       | MDNS      | 102    | Standard query 0x740e PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...  |
| 18  | 0.166058581 | 10.20.59.134           | 224.0.0.251       | MDNS      | 102    | Standard query 0x6305 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...  |
| 19  | 0.166058846 | 10.20.59.134           | 224.0.0.251       | MDNS      | 102    | Standard query 0xaec5 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...  |
| 20  | 0.201880956 | AzureWaveTec_86:a1:... | Broadcast         | ARP       | 60     | who has 10.20.56.86? Tell 10.20.56.80                                 |
| 21  | 0.202592169 | AzureWaveTec_86:a1:... | Broadcast         | ARP       | 60     | who has 10.20.31.21? Tell 10.20.56.80                                 |
| 22  | 0.202592554 | AzureWaveTec_86:a1:... | Broadcast         | ARP       | 60     | who has 10.20.54.113? Tell 10.20.56.80                                |
| 23  | 0.203558027 | AzureWaveTec_86:a1:... | Broadcast         | ARP       | 60     | who has 10.20.51.14? Tell 10.20.56.80                                 |
| 24  | 0.293143341 | fe80::cc7d:fd78:bea... | ff02::1:3         | LLMNR     | 95     | Standard query 0x4110 AAAA desktop-p6q8s1l                            |
| 25  | 0.293143569 | fe80::cc7d:fd78:bea... | ff02::1:3         | LLMNR     | 95     | Standard query 0x3b60 A desktop-p6q8s1l                               |
| 26  | 0.293143587 | 10.20.51.218           | 224.0.0.252       | LLMNR     | 75     | Standard query 0x3b60 A desktop-p6q8s1l                               |

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0  
 Ethernet II, Src: AzureWaveTec\_2c:6b:a0 (1c:ce:51:2c:6b:a0), Dst: Broadcast (ff:ff:ff:ff:ff:ff)  
 Address Resolution Protocol (request)



# DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 6<sup>th</sup> | Practical Assignment | Cyber Security (23010E004)

Date: 23/02/25

| No.                                                                                             | Time        | Source                 | Destination       | Protocol    | Length | Info                                                                  |
|-------------------------------------------------------------------------------------------------|-------------|------------------------|-------------------|-------------|--------|-----------------------------------------------------------------------|
| 7                                                                                               | 0.939712226 | GrandstreamN_15:0b:... | Broadcast         | Ethernet II | 628    | Ethernet II                                                           |
| 8                                                                                               | 0.953601703 | fe80::1cf0:d8eb:706... | ff02::1:ffb5:2e12 | ICMPv6      | 86     | Neighbor Solicitation for fe80::8fc:407b:48b5:2e12 from 3c:a6:f6:...  |
| 9                                                                                               | 0.953602587 | fe80::1cf0:d8eb:706... | ff02::1:ffc0:8864 | ICMPv6      | 86     | Neighbor Solicitation for fe80::a8ab:b5ff:fec0:8864 from 3c:a6:f6:... |
| 10                                                                                              | 0.953602606 | fe80::1cf0:d8eb:706... | ff02::1:ff30:166f | ICMPv6      | 86     | Neighbor Solicitation for fe80::1480:61c9:ae30:166f from 3c:a6:f6:... |
| 11                                                                                              | 0.953602624 | fe80::1cf0:d8eb:706... | ff02::1:ffea:76c5 | ICMPv6      | 86     | Neighbor Solicitation for fe80::1cfa:6a:42ea:76c5 from 3c:a6:f6:4...  |
| 12                                                                                              | 0.953602643 | fe80::1cf0:d8eb:706... | ff02::1:ffdb:dc03 | ICMPv6      | 86     | Neighbor Solicitation for fe80::c3e:6bf:10db:dc03 from 3c:a6:f6:4...  |
| 13                                                                                              | 0.953648644 | fe80::1cf0:d8eb:706... | ff02::1:ffb7:711f | ICMPv6      | 86     | Neighbor Solicitation for fe80::1c2a:cbal:f0b7:711f from 3c:a6:f6:... |
| 14                                                                                              | 0.161781104 | 10.20.59.65            | 224.0.0.251       | MDNS        | 102    | Standard query 0xc076 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...  |
| 15                                                                                              | 0.163026374 | 10.20.59.65            | 224.0.0.251       | MDNS        | 102    | Standard query 0x95d9 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...  |
| 16                                                                                              | 0.163026450 | 10.20.59.65            | 224.0.0.251       | MDNS        | 102    | Standard query 0x83b8 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...  |
| 17                                                                                              | 0.163965736 | 10.20.59.134           | 224.0.0.251       | MDNS        | 102    | Standard query 0x740e PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...  |
| 18                                                                                              | 0.166058581 | 10.20.59.134           | 224.0.0.251       | MDNS        | 102    | Standard query 0x6305 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...  |
| 19                                                                                              | 0.166058846 | 10.20.59.134           | 224.0.0.251       | MDNS        | 102    | Standard query 0xae5c PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...  |
| 20                                                                                              | 0.201880956 | AzureWaveTec_86:a1:... | Broadcast         | ARP         | 60     | Who has 10.20.56.86? Tell 10.20.56.80                                 |
| 21                                                                                              | 0.202592169 | AzureWaveTec_86:a1:... | Broadcast         | ARP         | 60     | Who has 10.20.31.21? Tell 10.20.56.80                                 |
| 22                                                                                              | 0.202592554 | AzureWaveTec_86:a1:... | Broadcast         | ARP         | 60     | Who has 10.20.54.113? Tell 10.20.56.80                                |
| 23                                                                                              | 0.203558027 | AzureWaveTec_86:a1:... | Broadcast         | ARP         | 60     | Who has 10.20.51.14? Tell 10.20.56.80                                 |
| 24                                                                                              | 0.293143341 | fe80::cc7d:fd78:bea... | ff02::1:1         | LLMNR       | 95     | Standard query 0x4110 AAAA desktop-p6q8s1l                            |
| 25                                                                                              | 0.293143569 | fe80::cc7d:fd78:bea... | ff02::1:1         | LLMNR       | 95     | Standard query 0x3b00 A desktop-p6q8s1l                               |
| 26                                                                                              | 0.293143587 | 10.20.51.218           | 224.0.0.252       | LLMNR       | 75     | Standard query 0x3b00 A desktop-p6q8s1l                               |
| 27                                                                                              | 0.293143604 | 10.20.51.218           | 224.0.0.252       | LLMNR       | 75     | Standard query 0x4110 AAAA desktop-p6q8s1l                            |
| 28                                                                                              | 0.303182711 | 60:ff:9e:1e:3c:6c      | Broadcast         | ARP         | 60     | Who has 10.20.38.17? Tell 10.20.59.196                                |
| 29                                                                                              | 0.303182943 | 60:ff:9e:1e:3c:6c      | Broadcast         | ARP         | 60     | Who has 10.20.38.13? Tell 10.20.59.196                                |
| 30                                                                                              | 0.309171238 | Intel_9d:c8:26         | Broadcast         | ARP         | 60     | Who has 169.254.169.254? Tell 10.20.50.181                            |
| 31                                                                                              | 0.378143077 | GrandstreamN_d4:50:... | Broadcast         | Ethernet II | 628    | Ethernet II                                                           |
| 32                                                                                              | 0.388995234 | 10.20.51.70            | 224.0.0.251       | MDNS        | 79     | Standard query 0x23f0 PTR arduino.tcp.local, "QM" question            |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0      |             |                        |                   |             |        |                                                                       |
| Ethernet II, Src: AzureWaveTec_2c:6b:a0 (1c:ce:51:2c:6b:a0), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |             |                        |                   |             |        |                                                                       |
| Address Resolution Protocol (request)                                                           |             |                        |                   |             |        |                                                                       |

| No.                                                                                             | Time         | Source                 | Destination       | Protocol    | Length | Info                                                                   |
|-------------------------------------------------------------------------------------------------|--------------|------------------------|-------------------|-------------|--------|------------------------------------------------------------------------|
| 44                                                                                              | 0.751056113  | AzureWaveTec_86:a1:... | Broadcast         | ARP         | 60     | Who has 10.20.31.21? Tell 10.20.56.80                                  |
| 45                                                                                              | 0.811491870  | fe80::811:d4aa:5591... | ff02::1:1         | LLMNR       | 88     | Standard query 0x1405 A 05_18_04                                       |
| 46                                                                                              | 0.811492107  | 10.20.28.4             | 224.0.0.252       | LLMNR       | 88     | Standard query 0x1405 A 05_18_04                                       |
| 47                                                                                              | 0.811492124  | fe80::811:d4aa:5591... | ff02::1:1         | LLMNR       | 88     | Standard query 0x1cf4 AAAA 05_18_04                                    |
| 48                                                                                              | 0.811492143  | 10.20.28.4             | 224.0.0.252       | LLMNR       | 88     | Standard query 0x1cf4 AAAA 05_18_04                                    |
| 49                                                                                              | 0.862474925  | fe80::2ea2:9cae:571... | ff02::1:ff21:a03  | ICMPv6      | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:...   |
| 50                                                                                              | 0.879657854  | AzureWaveTec_5c:7c:... | Broadcast         | ARP         | 60     | Who has 169.254.169.254? Tell 10.20.50.50                              |
| 51                                                                                              | 0.899330883  | AzureWaveTec_2c:6b:... | Broadcast         | ARP         | 60     | Who has 10.20.53.218? Tell 10.20.45.32                                 |
| 52                                                                                              | 1.031414661  | GrandstreamN_15:0b:... | Broadcast         | Ethernet II | 628    | Ethernet II                                                            |
| 53                                                                                              | 1.031943026  | GrandstreamN_15:0b:... | Broadcast         | Ethernet II | 628    | Ethernet II                                                            |
| 54                                                                                              | 1.031943292  | GrandstreamN_15:0b:... | Broadcast         | Ethernet II | 628    | Ethernet II                                                            |
| 55                                                                                              | 1.065915483  | fe80::1cf0:d8eb:706... | ff02::1:ff30:166f | ICMPv6      | 86     | Neighbor Solicitation for fe80::1480:61c9:ae30:166f from 3c:a6:f6:...  |
| 56                                                                                              | 1.065915713  | fe80::1cf0:d8eb:706... | ff02::1:ffb5:2e12 | ICMPv6      | 86     | Neighbor Solicitation for fe80::8fc:407b:48b5:2e12 from 3c:a6:f6:...   |
| 57                                                                                              | 1.065915731  | fe80::1cf0:d8eb:706... | ff02::1:ffea:76c5 | ICMPv6      | 86     | Neighbor Solicitation for fe80::1cfa:6a:42ea:76c5 from 3c:a6:f6:4...   |
| 58                                                                                              | 1.065915747  | fe80::1cf0:d8eb:706... | ff02::1:ffc0:8864 | ICMPv6      | 86     | Neighbor Solicitation for fe80::a8ab:b5ff:fec0:8864 from 3c:a6:f6:4... |
| 59                                                                                              | 1.065915765  | fe80::1cf0:d8eb:706... | ff02::1:ffdb:dc03 | ICMPv6      | 86     | Neighbor Solicitation for fe80::c3e:6bf:10db:dc03 from 3c:a6:f6:4...   |
| 60                                                                                              | 1.065915782  | fe80::1cf0:d8eb:706... | ff02::1:ffb7:711f | ICMPv6      | 86     | Neighbor Solicitation for fe80::1c2a:cbal:f0b7:711f from 3c:a6:f6:...  |
| 61                                                                                              | 1.086599448  | 10.20.51.70            | 224.0.0.251       | MDNS        | 102    | Standard query 0xec5f PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...   |
| 62                                                                                              | 1.086599675  | 10.20.51.70            | 224.0.0.251       | MDNS        | 102    | Standard query 0xc374 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...   |
| 63                                                                                              | 1.087407223  | 10.20.59.65            | 224.0.0.251       | MDNS        | 102    | Standard query 0xe6cc PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...   |
| 64                                                                                              | 1.087407448  | 10.20.59.65            | 224.0.0.251       | MDNS        | 102    | Standard query 0x29c4 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...   |
| 65                                                                                              | 1.0924087935 | 10.20.59.134           | 224.0.0.251       | MDNS        | 102    | Standard query 0x5560 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo...   |
| 66                                                                                              | 1.092408178  | 10.20.59.134           | 224.0.0.251       | MDNS        | 102    | Standard query 0x134d PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo...   |
| 67                                                                                              | 1.193078577  | 10.20.59.65            | 224.0.0.251       | MDNS        | 85     | Standard query 0x21aa PTR Bhautik.dosvc.tcp.local, "QM" question       |
| 68                                                                                              | 1.193078807  | 10.20.59.65            | 224.0.0.251       | MDNS        | 102    | Standard query 0x7bd2 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo...   |
| 69                                                                                              | 1.195732300  | 10.20.59.134           | 224.0.0.251       | MDNS        | 85     | Standard query 0xdeb9 PTR Bhautik.dosvc.tcp.local, "QM" question       |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0      |              |                        |                   |             |        |                                                                        |
| Ethernet II, Src: AzureWaveTec_2c:6b:a0 (1c:ce:51:2c:6b:a0), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |              |                        |                   |             |        |                                                                        |
| Address Resolution Protocol (request)                                                           |              |                        |                   |             |        |                                                                        |

| No.                                                                                             | Time        | Source                 | Destination            | Protocol    | Length | Info                                                               |
|-------------------------------------------------------------------------------------------------|-------------|------------------------|------------------------|-------------|--------|--------------------------------------------------------------------|
| 80                                                                                              | 1.434481061 | GrandstreamN_3d:1b:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| 81                                                                                              | 1.434481289 | GrandstreamN_3d:1b:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| 82                                                                                              | 1.438563323 | GrandstreamN_d4:50:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| 83                                                                                              | 1.444740273 | AzureWaveTec_5c:7c:... | Broadcast              | ARP         | 60     | Who has 169.254.169.254? Tell 10.20.50.50                          |
| 84                                                                                              | 1.449739751 | GrandstreamN_d4:50:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| 85                                                                                              | 1.575882992 | AzureWaveTec_2c:6b:... | Broadcast              | ARP         | 60     | Who has 10.20.43.30? Tell 10.20.45.23                              |
| 86                                                                                              | 1.638560268 | GrandstreamN_15:0b:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| 87                                                                                              | 1.732437336 | Sophos_ce:2f:57        | Broadcast              | ARP         | 60     | Who has 10.20.45.27? Tell 10.20.1.1                                |
| 88                                                                                              | 1.737015115 | LiteonTechno_39:ae:... | Broadcast              | LLC         | 60     | S P; func=RN; N(R)=64; DSAP NULL LSAP Individual, SSAP NULL LSAP   |
| 89                                                                                              | 1.745304134 | LiteonTechno_39:ae:... | GrandstreamN_d4:50:... | EAPOL       | 135    | Key (Message 2 of 4)                                               |
| 90                                                                                              | 1.751360422 | LiteonTechno_39:ae:... | GrandstreamN_d4:50:... | EAPOL       | 113    | Key (Message 4 of 4)                                               |
| 91                                                                                              | 1.757634391 | AzureWaveTec_86:a1:... | Broadcast              | ARP         | 60     | Who has 10.20.54.113? Tell 10.20.56.80                             |
| 92                                                                                              | 1.757634616 | AzureWaveTec_86:a1:... | Broadcast              | ARP         | 60     | Who has 10.20.56.86? Tell 10.20.56.80                              |
| 93                                                                                              | 1.757634633 | AzureWaveTec_86:a1:... | Broadcast              | ARP         | 60     | Who has 10.20.31.21? Tell 10.20.56.80                              |
| 94                                                                                              | 1.757634652 | AzureWaveTec_86:a1:... | Broadcast              | ARP         | 60     | Who has 10.20.51.14? Tell 10.20.56.80                              |
| 95                                                                                              | 1.910093926 | fe80::8379:4a63:ed3... | ff02::1:1              | LLMNR       | 95     | Standard query 0x954c ANY LAPTOP-UURSOCJ2                          |
| 96                                                                                              | 1.910093979 | fe80::8379:4a63:ed3... | ff02::1:2              | DHCPv6      | 120    | Information-request XID: 0x694e2 CID: 000100012e96b33800010017f... |
| 97                                                                                              | 1.910570907 | 0.0.0.0                | 255.255.255.255        | DHCP        | 364    | DHCP Request - Transaction ID 0xf18971b4                           |
| 98                                                                                              | 1.910570924 | 10.20.57.126           | 224.0.0.252            | LLMNR       | 75     | Standard query 0x954c ANY LAPTOP-UURSOCJ2                          |
| 99                                                                                              | 1.916020445 | AzureWaveTec_2c:6b:... | Broadcast              | ARP         | 60     | Who has 10.20.53.218? Tell 10.20.45.32                             |
| 100                                                                                             | 1.950598010 | LiteonTechno_39:ae:... | Broadcast              | ARP         | 60     | Who has 10.20.1.1? Tell 10.20.57.126                               |
| 101                                                                                             | 1.982115530 | fe80::8379:4a63:ed3... | ff02::2                | ICMPv6      | 62     | Router Solicitation                                                |
| 102                                                                                             | 1.982115799 | ::                     | ff02::1:ff3e:9ecc      | ICMPv6      | 78     | Neighbor Solicitation for fe80::8379:4a63:ed3e:9ecc                |
| 103                                                                                             | 1.982115817 | LiteonTechno_39:ae:... | Broadcast              | ARP         | 60     | Who has 10.20.57.126? (ARP Probe)                                  |
| 104                                                                                             | 2.032827168 | GrandstreamN_15:0b:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| 105                                                                                             | 2.032827393 | GrandstreamN_15:0b:... | Broadcast              | Ethernet II | 628    | Ethernet II                                                        |
| Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0      |             |                        |                        |             |        |                                                                    |
| Ethernet II, Src: AzureWaveTec_2c:6b:a0 (1c:ce:51:2c:6b:a0), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |             |                        |                        |             |        |                                                                    |
| Address Resolution Protocol (request)                                                           |             |                        |                        |             |        |                                                                    |



Date: 23/02/25

### 3. Detecting Man-in-the-Middle (MITM) Attacks

#### a. Monitor ARP poisoning attempts

| No. | Time        | Source                 | Destination            | Protocol  | Length | Info                                                                 |
|-----|-------------|------------------------|------------------------|-----------|--------|----------------------------------------------------------------------|
| 1   | 0.000000000 | GrandstreamN_15:0b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 2   | 0.000067005 | GrandstreamN_15:0b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 3   | 0.043008825 | fe80::e3b9:9d38:615... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:... |
| 4   | 0.052539186 | ChongqingFug_78:28...  | Broadcast              | ARP       | 60     | Who has 10.20.60.103? Tell 10.20.53.191                              |
| 5   | 0.052539417 | ChongqingFug_78:28...  | Broadcast              | ARP       | 60     | Who has 10.20.51.72? Tell 10.20.53.191                               |
| 6   | 0.101416410 | 10.20.53.250           | 224.0.0.251            | MDNS      | 85     | Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question   |
| 7   | 0.132102526 | AzureWaveTec_2b:ce...  | Broadcast              | ARP       | 60     | Who has 10.20.58.73? Tell 10.20.45.29                                |
| 8   | 0.132102800 | AzureWaveTec_2b:ce...  | Broadcast              | ARP       | 60     | Who has 10.20.38.13? Tell 10.20.45.29                                |
| 9   | 0.154313190 | GrandstreamN_d4:50...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 10  | 0.197176615 | GrandstreamN_3d:1b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 11  | 0.206757120 | GrandstreamN_d4:50...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 12  | 0.532288606 | AzureWaveTec_d9:14...  | Broadcast              | ARP       | 60     | Who has 10.20.56.39? Tell 10.20.55.37                                |
| 13  | 0.532288835 | AzureWaveTec_d9:14...  | Broadcast              | ARP       | 60     | Who has 10.20.50.202? Tell 10.20.55.37                               |
| 14  | 0.558901753 | 10.20.60.19            | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 15  | 0.560055408 | 10.20.60.14            | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 16  | 0.562264417 | 10.20.60.13            | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 17  | 0.625388552 | fe80::e3b9:9d38:615... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:... |
| 18  | 0.679777732 | Intel_ab:b2:79         | Broadcast              | ARP       | 60     | Who has 10.20.53.219? Tell 10.20.53.236                              |
| 19  | 0.694647884 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.26.46? Tell 10.20.1.1                                  |
| 20  | 0.821929953 | DLink_7c:96:42         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 21  | 0.978545522 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.54.145? Tell 10.20.1.1                                 |
| 22  | 0.984244843 | 10.20.51.70            | 224.0.0.251            | MDNS      | 97     | Standard query 0x6bbf PTR Malvi's MacBook Air._rfb._tcp.local, "Q... |
| 23  | 1.001232453 | GrandstreamN_15:0b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 24  | 1.025848363 | GrandstreamN_15:0b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 25  | 1.055906506 | DLink_b8:a2:a0         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 26  | 1.107455041 | AzureWaveTec_2c:68...  | Broadcast              | ARP       | 60     | Who has 10.20.37.2? Tell 10.20.45.19                                 |

Frame 1: 628 bytes on wire (5024 bits), 628 bytes captured (5024 bits) on interface eth0, id 0  
 Ethernet II, Src: GrandstreamN\_15:0b:ec (ec:74:d7:15:0b:ec), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

| No. | Time        | Source                 | Destination            | Protocol  | Length | Info                                                                 |
|-----|-------------|------------------------|------------------------|-----------|--------|----------------------------------------------------------------------|
| 22  | 0.984244843 | 10.20.51.70            | 224.0.0.251            | MDNS      | 97     | Standard query 0x6bbf PTR Malvi's MacBook Air._rfb._tcp.local, "Q... |
| 23  | 1.001232453 | GrandstreamN_15:0b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 24  | 1.025848363 | GrandstreamN_15:0b...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 25  | 1.055906506 | DLink_b8:a2:a0         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 26  | 1.107455041 | AzureWaveTec_2c:68...  | Broadcast              | ARP       | 60     | Who has 10.20.37.2? Tell 10.20.45.19                                 |
| 27  | 1.130693010 | AzureWaveTec_2b:ce...  | Broadcast              | ARP       | 60     | Who has 10.20.58.73? Tell 10.20.45.29                                |
| 28  | 1.130693256 | AzureWaveTec_2b:ce...  | Broadcast              | ARP       | 60     | Who has 10.20.38.13? Tell 10.20.45.29                                |
| 29  | 1.146645763 | fe80::1fe1:86e6:968... | ff02::1:ff9f:d9cd      | ICMPv6    | 86     | Neighbor Solicitation for fe80::2423:26ce:8c9f:d9cd from b8:86:87... |
| 30  | 1.357383361 | 10.20.60.21            | 255.255.255.255        | UDP       | 932    | 49832 → 29810 Len=890                                                |
| 31  | 1.374779709 | Intel_91:ec:d0         | Broadcast              | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.52.144                                 |
| 32  | 1.536913594 | AzureWaveTec_d9:14...  | Broadcast              | ARP       | 60     | Who has 10.20.56.39? Tell 10.20.55.37                                |
| 33  | 1.536913830 | AzureWaveTec_d9:14...  | Broadcast              | ARP       | 60     | Who has 10.20.50.202? Tell 10.20.55.37                               |
| 34  | 1.623796763 | fe80::e3b9:9d38:615... | ff02::1:ff21:a03       | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from 1c:ce:51:... |
| 35  | 1.737758862 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.44.19? Tell 10.20.1.1                                  |
| 36  | 1.768236536 | AzureWaveTec_2c:68...  | Broadcast              | ARP       | 60     | Who has 10.20.37.2? Tell 10.20.45.19                                 |
| 37  | 1.835177807 | fe80::811:d4aa:5591... | ff02::1:3              | LLMNR     | 88     | Standard query 0xc69b A 05_18_04                                     |
| 38  | 1.835206721 | 10.20.28.4             | 224.0.0.252            | LLMNR     | 68     | Standard query 0xc69b A 05_18_04                                     |
| 39  | 1.835614012 | fe80::811:d4aa:5591... | ff02::1:3              | LLMNR     | 88     | Standard query 0xbde5 AAAA 05_18_04                                  |
| 40  | 1.835614085 | 10.20.28.4             | 224.0.0.252            | LLMNR     | 68     | Standard query 0xbde5 AAAA 05_18_04                                  |
| 41  | 1.951803831 | fe80::1fe1:86e6:968... | ff02::1:ff9f:d9cd      | ICMPv6    | 86     | Neighbor Solicitation for fe80::2423:26ce:8c9f:d9cd from b8:86:87... |
| 42  | 2.002547845 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.54.145? Tell 10.20.1.1                                 |
| 43  | 2.008821218 | 10.20.51.70            | 224.0.0.251            | MDNS      | 97     | Standard query 0x28a2 PTR Malvi's MacBook Air._rfb._tcp.local, "Q... |
| 44  | 2.126474821 | 8a:38:57:06:4a:52      | Broadcast              | ARP       | 60     | Who has 10.20.57.130? Tell 10.20.42.27                               |
| 45  | 2.136338377 | ChongqingFug_78:28...  | Broadcast              | ARP       | 60     | Who has 10.20.60.103? Tell 10.20.53.191                              |
| 46  | 2.136338789 | ChongqingFug_78:28...  | Broadcast              | ARP       | 60     | Who has 10.20.51.72? Tell 10.20.53.191                               |
| 47  | 2.244990570 | GrandstreamN_d4:50...  | Broadcast              | Ethern... | 628    | Ethernet II                                                          |

Frame 1: 628 bytes on wire (5024 bits), 628 bytes captured (5024 bits) on interface eth0, id 0  
 Ethernet II, Src: GrandstreamN\_15:0b:ec (ec:74:d7:15:0b:ec), Dst: Broadcast (ff:ff:ff:ff:ff:ff)





# DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 6<sup>th</sup> | Practical Assignment | Cyber Security (23010E004)

Date: 23/02/25

| No.                                                                                             | Time        | Source                 | Destination            | Protocol  | Length | Info                                                                 |
|-------------------------------------------------------------------------------------------------|-------------|------------------------|------------------------|-----------|--------|----------------------------------------------------------------------|
| 55                                                                                              | 2.556777046 | 10.20.53.18            | 224.0.0.251            | MDNS      | 267    | Standard query response 0x0000 PTR, cache flush Maulik-M2s-MacBoo... |
| 56                                                                                              | 2.556777275 | 10.20.53.18            | 224.0.0.251            | MDNS      | 324    | Standard query 0x0000 PTR lb._dns-sd._udp.local, "QU" question PT... |
| 57                                                                                              | 2.561906457 | ::                     | ff02::1:ffe7:4c0a      | ICMPv6    | 86     | Neighbor Solicitation for fe80::1466:6d63:1be7:4c0a                  |
| 58                                                                                              | 2.562440140 | 0.0.0.0                | 255.255.255.255        | DHCP      | 342    | DHCP Request - Transaction ID 0xdb9c506                              |
| 59                                                                                              | 2.562440365 | fe80::1466:6d63:1be... | ff02::2                | ICMPv6    | 62     | Router Solicitation                                                  |
| 60                                                                                              | 2.564786468 | GoodWayInd_af:ae:a0    | Broadcast              | ARP       | 60     | ARP Announcement for 10.20.53.18                                     |
| 61                                                                                              | 2.567014934 | GoodWayInd_af:ae:a0    | Broadcast              | ARP       | 60     | Who has 169.254.255.255? Tell 10.20.53.18                            |
| 62                                                                                              | 2.598547659 | GoodWayInd_af:ae:a0    | Broadcast              | ARP       | 60     | Who has 10.20.1.1? Tell 10.20.53.18                                  |
| 63                                                                                              | 2.599387270 | 10.20.53.18            | 224.0.0.251            | MDNS      | 267    | Standard query response 0x0000 PTR, cache flush Maulik-M2s-MacBoo... |
| 64                                                                                              | 2.642796687 | 10.20.53.18            | 224.0.0.251            | MDNS      | 324    | Standard query 0x0000 PTR lb._dns-sd._udp.local, "QU" question PT... |
| 65                                                                                              | 2.679052531 | AzureWaveTec_d9:14:... | Broadcast              | ARP       | 60     | Who has 10.20.50.202? Tell 10.20.55.37                               |
| 66                                                                                              | 2.679052764 | AzureWaveTec_d9:14:... | Broadcast              | ARP       | 60     | Who has 10.20.56.39? Tell 10.20.55.37                                |
| 67                                                                                              | 2.685910879 | 8a:38:57:66:4a:52      | Broadcast              | ARP       | 60     | Who has 10.20.57.130? Tell 10.20.42.27                               |
| 68                                                                                              | 2.739749130 | Sophos_ce:2f:57        | Broadcast              | ARP       | 60     | Who has 10.20.44.19? Tell 10.20.1.1                                  |
| 69                                                                                              | 2.756934812 | AzureWaveTec_2c:68:... | Broadcast              | ARP       | 60     | Who has 10.20.37.2? Tell 10.20.45.19                                 |
| 70                                                                                              | 2.804762662 | 10.20.60.20            | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 71                                                                                              | 2.806782428 | 10.20.60.6             | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 72                                                                                              | 2.808203633 | 10.20.60.18            | 224.0.0.120            | IPv4      | 208    | Unassigned (153)                                                     |
| 73                                                                                              | 2.821906321 | DLink_7c:96:20         | Ethernet-Configurat... | LOOP      | 60     | Unknown function (256)                                               |
| 74                                                                                              | 2.888337790 | GoodWayInd_af:ae:a0    | Broadcast              | ARP       | 60     | Who has 169.254.255.255? Tell 10.20.53.18                            |
| 75                                                                                              | 2.923924897 | GrandstreamN_d4:50:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 76                                                                                              | 2.923924717 | GrandstreamN_d4:50:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 77                                                                                              | 2.938452866 | fe80::1fe1:86e6:908... | ff02::1:ff9f:d9cd      | ICMPv6    | 86     | Neighbor Solicitation for fe80::2423:26ce:8c9f:d9cd from b8:86:87... |
| 78                                                                                              | 2.966536831 | Intel_lab:b2:79        | Broadcast              | ARP       | 60     | Who has 10.20.53.219? Tell 10.20.53.236                              |
| 79                                                                                              | 3.003815407 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| 80                                                                                              | 3.003815698 | GrandstreamN_15:0b:... | Broadcast              | Ethern... | 628    | Ethernet II                                                          |
| Frame 1: 628 bytes on wire (5024 bits), 628 bytes captured (5024 bits) on interface eth0, id 0  |             |                        |                        |           |        |                                                                      |
| Ethernet II, Src: GrandstreamN_15:0b:ec (ec:74:d7:15:0b:ec), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |             |                        |                        |           |        |                                                                      |

| No.                                                                                             | Time         | Source                 | Destination      | Protocol  | Length | Info                                                                 |
|-------------------------------------------------------------------------------------------------|--------------|------------------------|------------------|-----------|--------|----------------------------------------------------------------------|
| 340                                                                                             | 9.283211881  | 10.20.59.139           | 224.0.0.251      | MDNS      | 102    | Standard query 0x4f91 PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 341                                                                                             | 9.283726227  | 10.20.59.180           | 224.0.0.251      | MDNS      | 102    | Standard query 0xe82d PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 342                                                                                             | 9.286491136  | 10.20.59.232           | 224.0.0.251      | MDNS      | 102    | Standard query 0x9b24 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 343                                                                                             | 9.286491365  | 10.20.59.232           | 224.0.0.251      | MDNS      | 102    | Standard query 0x1fe9 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 344                                                                                             | 9.287030896  | 10.20.59.232           | 224.0.0.251      | MDNS      | 102    | Standard query 0xa4ed PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 345                                                                                             | 9.291832602  | HP_6a:97:4d            | Broadcast        | ARP       | 60     | Who has 10.20.15.15? Tell 10.20.53.193                               |
| 346                                                                                             | 9.309907606  | 10.20.59.137           | 224.0.0.251      | MDNS      | 102    | Standard query 0x7779 PTR eed53bf1-b007-45b5-a497-116cc9e0631e.lo... |
| 347                                                                                             | 9.309907833  | 10.20.59.137           | 224.0.0.251      | MDNS      | 102    | Standard query 0x68e3 PTR a3b75187-5a5f-483a-b879-48f9ad7a25a8.lo... |
| 348                                                                                             | 9.310101773  | 10.20.59.137           | 224.0.0.251      | MDNS      | 102    | Standard query 0x5e7b PTR 5da4cb9b-cafd-44d1-b5ae-5a4cb25bd87b.lo... |
| 349                                                                                             | 9.410738005  | 10.20.50.99            | 224.0.0.251      | MDNS      | 391    | Standard query response 0x0000 PTR, cache flush HP_11_dovsc_.tcp...  |
| 350                                                                                             | 9.410738273  | 10.20.50.99            | 224.0.0.251      | MDNS      | 337    | Standard query response 0x0000 SRV, cache flush 0 0 7680 HP_11.lo... |
| 351                                                                                             | 9.501705276  | 10.20.53.18            | 224.0.0.251      | MDNS      | 1145   | Standard query response 0x0000 TXT, cache flush PTR _airplay_.tcp... |
| 352                                                                                             | 9.534738625  | GrandstreamN_15:0b:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 353                                                                                             | 9.551983592  | fe80::87c4:da3b:2c5... | ff02::1:ff21:a03 | ICMPv6    | 86     | Neighbor Solicitation for fe80::bd79:cdcc:ed21:a03 from b4:8c:9d:... |
| 354                                                                                             | 9.556301246  | AzureWaveTec_a4:30:... | Broadcast        | ARP       | 60     | Who has 10.20.56.81? Tell 10.20.58.63                                |
| 355                                                                                             | 9.566652970  | fe80::604a:5030:eab... | ff02::c          | UDP       | 718    | 65322 → 3702 Len=656                                                 |
| 356                                                                                             | 9.573149626  | AzureWaveTec_86:a1:... | Broadcast        | ARP       | 60     | Who has 10.20.32.14? Tell 10.20.56.80                                |
| 357                                                                                             | 9.573149853  | AzureWaveTec_86:a1:... | Broadcast        | ARP       | 60     | Who has 10.20.43.39? Tell 10.20.56.80                                |
| 358                                                                                             | 9.573149870  | AzureWaveTec_86:a1:... | Broadcast        | ARP       | 60     | Who has 10.20.56.81? Tell 10.20.56.80                                |
| 359                                                                                             | 9.573149888  | AzureWaveTec_86:a1:... | Broadcast        | ARP       | 60     | Who has 10.20.32.9? Tell 10.20.56.80                                 |
| 360                                                                                             | 9.580962260  | GrandstreamN_d4:50:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| 361                                                                                             | 9.733886995  | AzureWaveTec_2c:6b:... | Broadcast        | ARP       | 60     | Who has 10.20.31.16? Tell 10.20.45.32                                |
| 362                                                                                             | 9.750264697  | 10.20.51.1             | 10.20.255.255    | NBNS      | 92     | Name query NB DESKTOP-02RVK6Q<20>                                    |
| 363                                                                                             | 9.757850497  | AzureWaveTec_2c:68:... | Broadcast        | ARP       | 60     | Who has 10.20.37.2? Tell 10.20.45.19                                 |
| 364                                                                                             | 9.782825404  | Sophos_ce:2f:57        | Broadcast        | ARP       | 60     | Who has 10.20.44.19? Tell 10.20.1.1                                  |
| 365                                                                                             | 10.031234573 | GrandstreamN_15:0b:... | Broadcast        | Ethern... | 628    | Ethernet II                                                          |
| Frame 1: 628 bytes on wire (5024 bits), 628 bytes captured (5024 bits) on interface eth0, id 0  |              |                        |                  |           |        |                                                                      |
| Ethernet II, Src: GrandstreamN_15:0b:ec (ec:74:d7:15:0b:ec), Dst: Broadcast (ff:ff:ff:ff:ff:ff) |              |                        |                  |           |        |                                                                      |